

Kapitel 9 – Perowskite



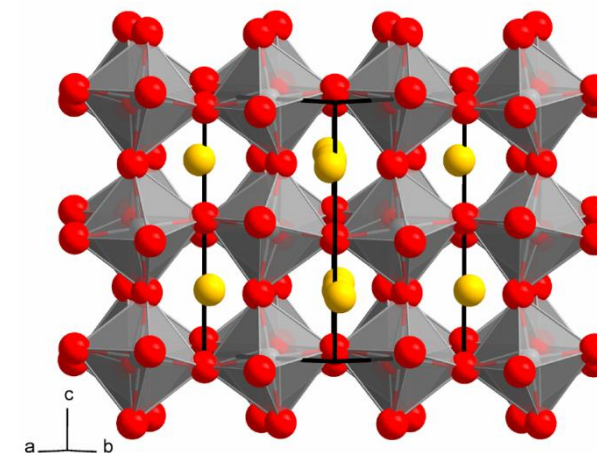
“Perowskit ist ein Mineral, **Calciumtitanat**, mit der Strukturformel CaTiO_3 .



Es wurde von Gustav Rose 1839 entdeckt und nach dem Russischen Mineralogen **Graf Lew Alexejewitsch von Perowski** (1792-1856) benannt



Heutzutage werden alle Mineralien mit derselben Perowskit-Struktur und chemischer Zusammensetzung, ABX_3 , Perowskite genannt



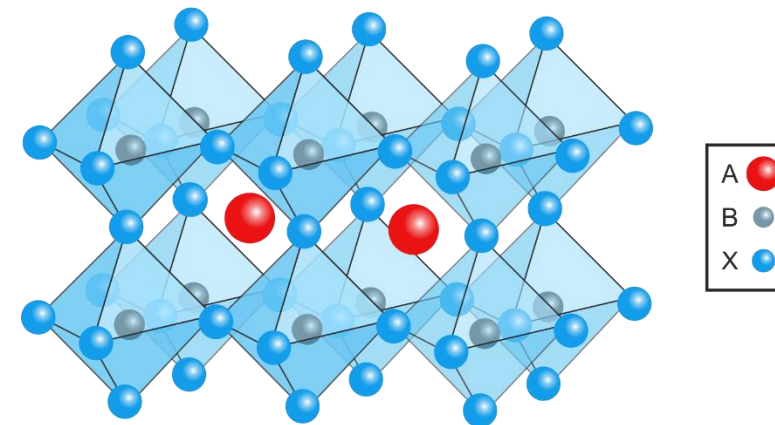
Zur Erhaltung der Ladungsneutralität gilt: $A + B = 3X$

1. Fall : $X = \text{II}$ (Oxide; $A+B = \text{VI}$)

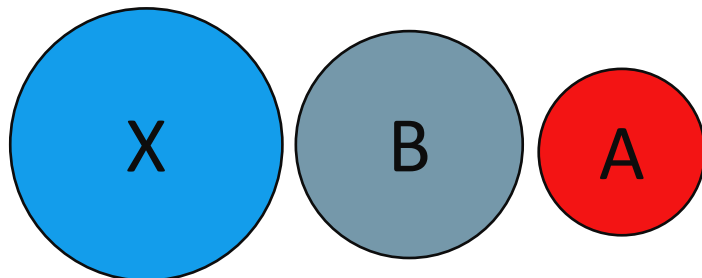
- $A = \text{I}, B = \text{V}$ (z.B. KTaO_3)
- $A = \text{II}, B = \text{IV}$ (z.B. SrTiO_3)
- $A = \text{III}, B = \text{III}$ (z.B. GdFeO_3)

2. Fall: $X = I$ (Halide; $A+B = III$)

- $A = \text{I}$, $B = \text{II}$ (z.B. CsSnI_3)



H																	He															
Li	Be									B	C	N	O	F	Ne																	
Na	Mg									Al	Si	P	S	Cl	Ar																	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr															
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe															
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn															
Fr	Ra																															
																		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
																		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr



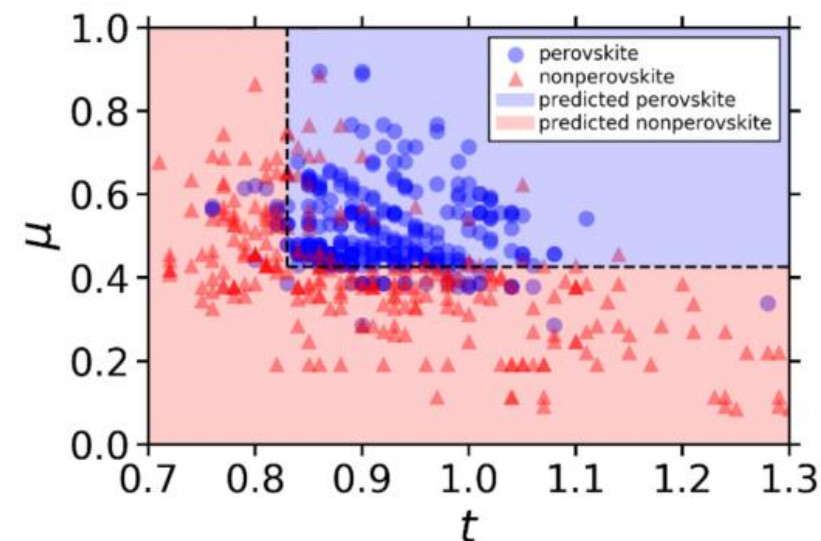
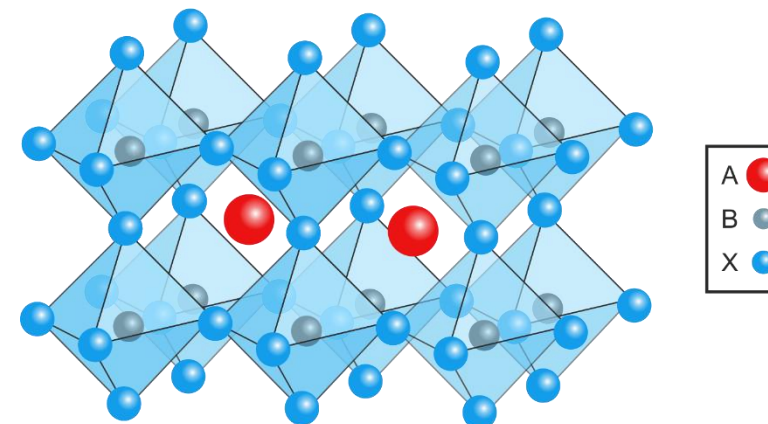
Die Radien der einzelnen Ionen in der Perowskit Struktur hängen voneinander ab.

→ Goldschmidt Toleranzfaktor

$$\alpha = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$$

→ Oktahedrfaktor

$$\mu = \frac{r_B}{r_X}$$



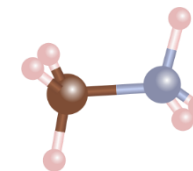
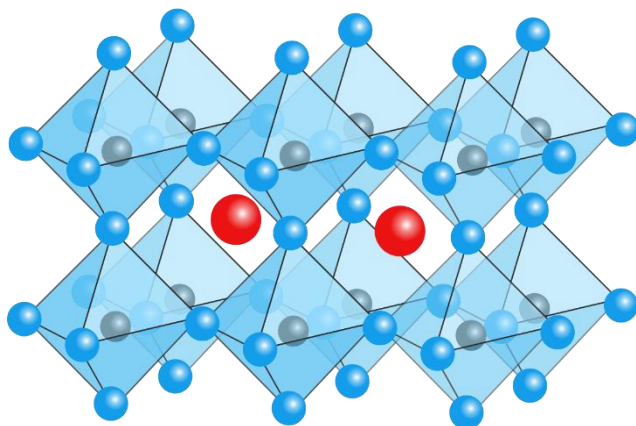
Hybride organisch/anorganische Perowskite

Isovalentes Ersetzen des A Kations: Statt eines Metallions ein organisches Kation:

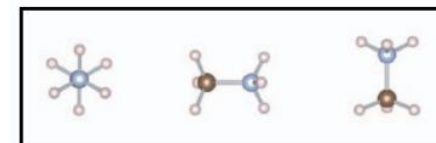
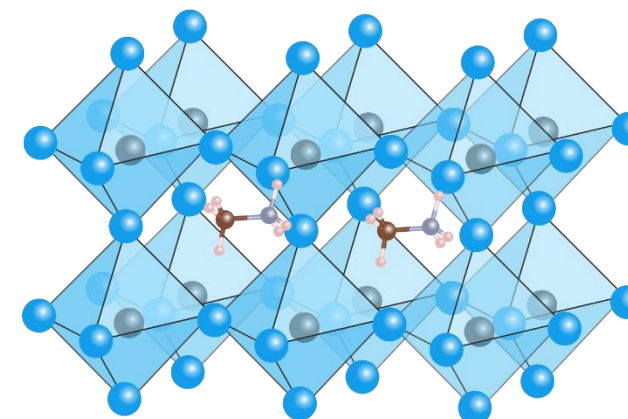
-> Hybride organisch/anorganische Perowskite



Caesium



Methylammonium (CH_3NH_3^+)



Variable Orientierung

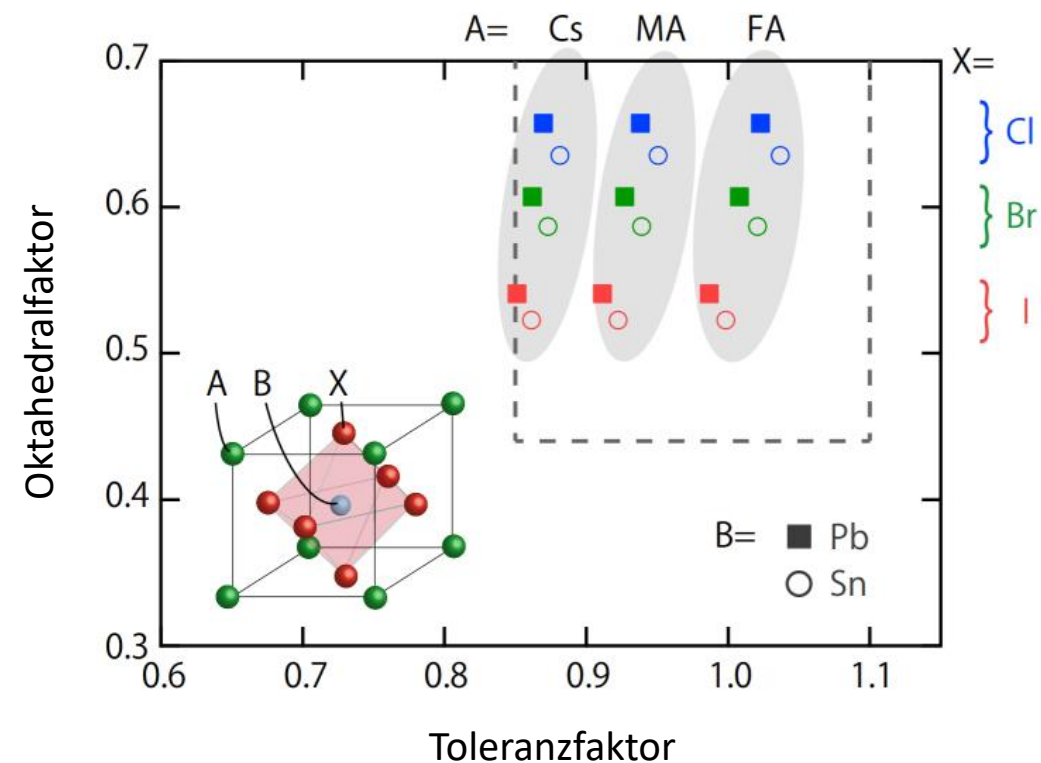
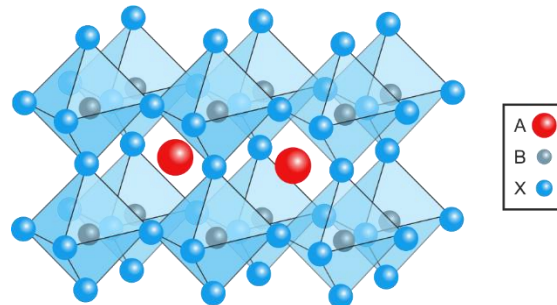
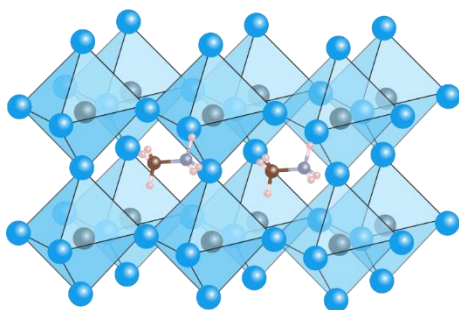
Häufigste A-Kationen

Goldschmidt Toleranzfaktor

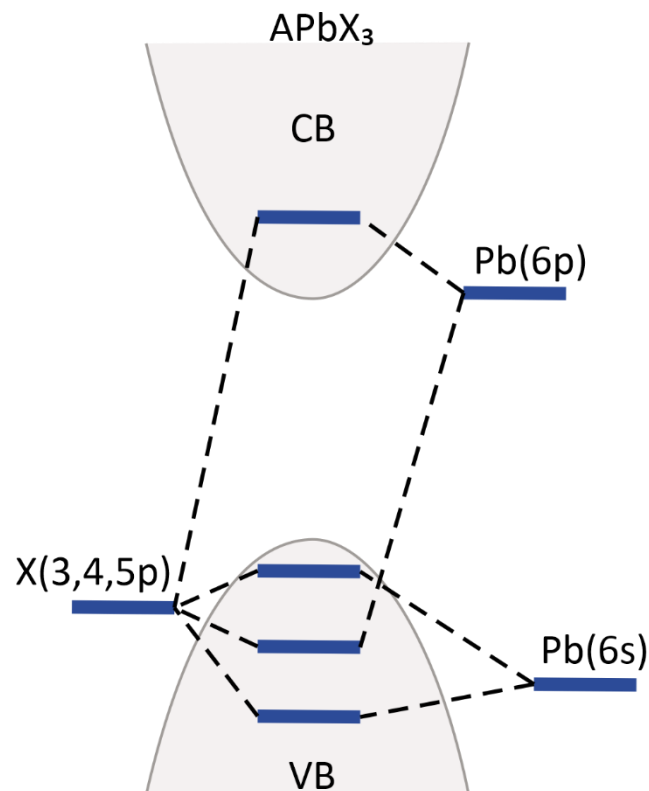
Oktahedalfaktor

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$$

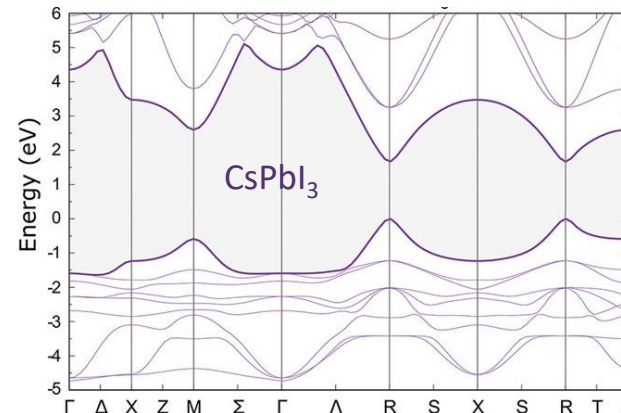
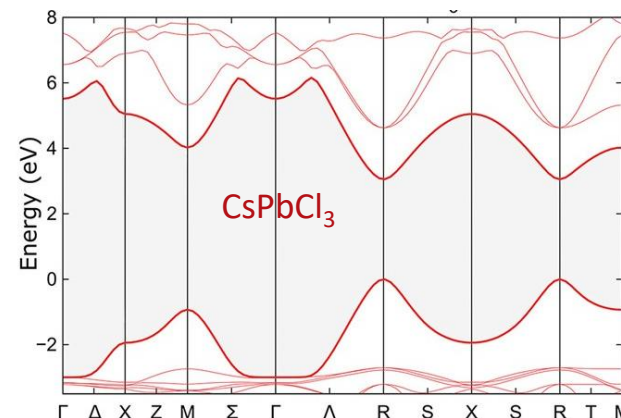
$$\mu = \frac{r_B}{r_X}$$



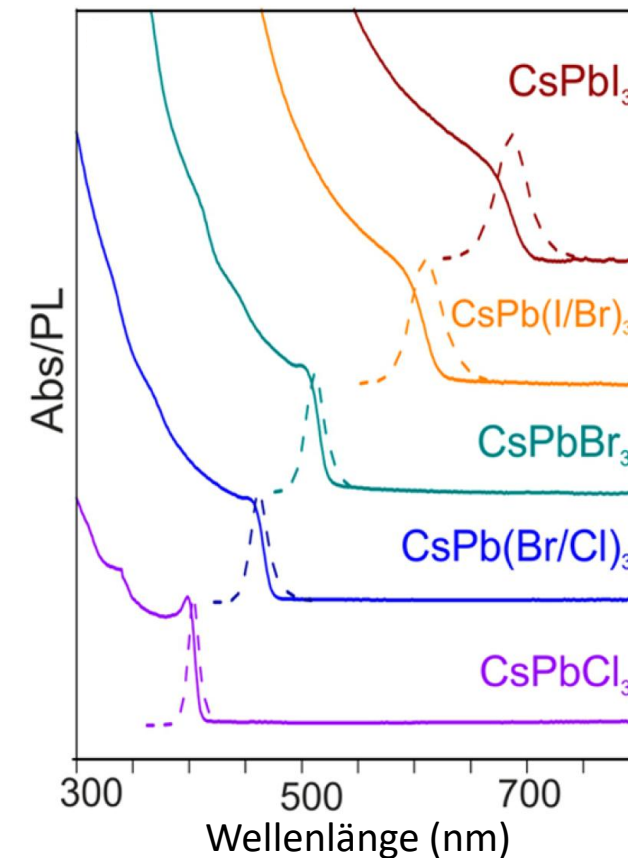
Die elektronische Bandstruktur wird hauptsächlich durch die Blei/Halid-Hybridisierung definiert



Die Bandlücke liegt zwischen zwei antibindenden Bändern



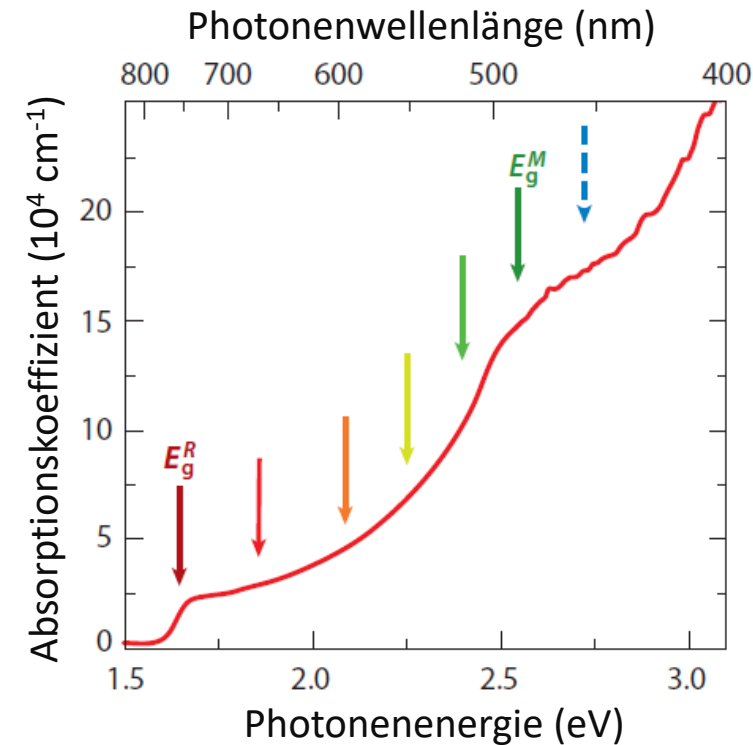
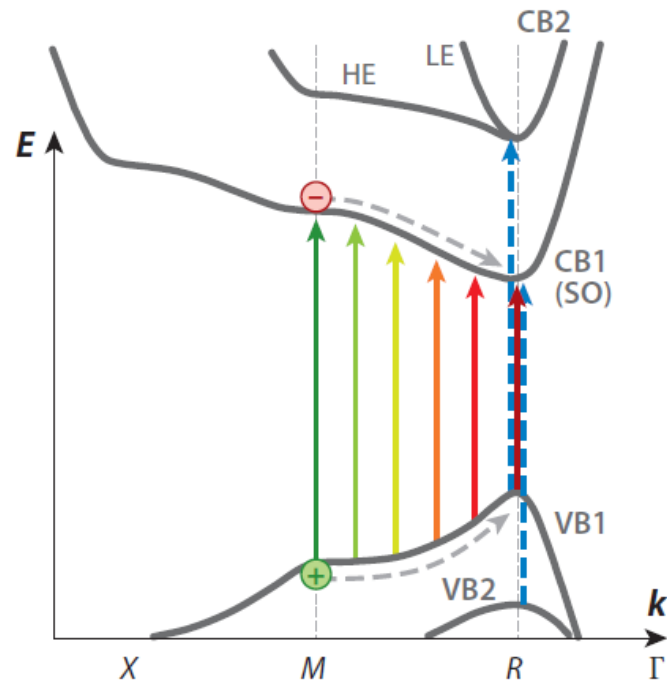
Leitungsband ist p-artig,
Valenzband ist s-artig



Bandlücke ist durch die
Halidzusammensetzung einstellbar

Warum sind Halidperowskite so gute Lichtabsorber ?

Wegen der Bandstruktur: Direkte Absorption entlang der gesamte M-R Linie möglich
→ Ladungsträger relaxieren alle zum R-Punkt



Halidperowskite: Ein alter Hund ... lernt neue Tricks

Erstes Paper

Über die Cäsium- und Kalium-Bleihalogenide.

Von
H. L. WELLS.¹

Als Fortsetzung der in diesem Laboratorium² begonnenen Arbeit über Doppelhalogenide ist von den Herren G. F. CAMPBELL, P. T. WALDEN und A. P. WHEELER eine Untersuchung über die Cäsium-Bleisalze unternommen worden. Diese Herren haben die Untersuchung mit vielem Eifer und Geschick durchgeführt, und es macht mir Freude, ihnen meinen Dank auszusprechen. Sie haben die Existenz folgender Salze konstatiert:

Cs_2PbCl_6	Cs_2PbBr_6	—
CsPbCl_3	CsPbBr_3	—
CsPb_2Cl_6	CsPb_2Br_6	—

$\text{CH}_3\text{NH}_3\text{PbX}_3$, ein Pb(II)-System mit kubischer Perowskitstruktur

$\text{CH}_3\text{NH}_3\text{PbX}_3$, a Pb(II)-System with Cubic Perovskite Structure

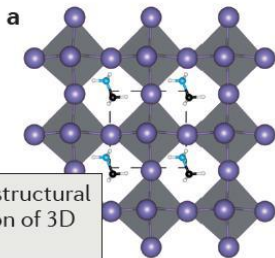
Dieter Weber

Institut für Anorganische Chemie der Universität Stuttgart

Z. Naturforsch. **33b**, 1443–1445 (1978); eingegangen am 21. August 1978

Synthesis, X-ray

$\text{CH}_3\text{NH}_3\text{PbX}_3$ (X = Cl, Br, I) has the cubic perovskite structure with the unit cell parameters $a = 5.68 \text{ \AA}$ (X = Cl), $a = 5.92 \text{ \AA}$ (X = Br) and $a = 6.27 \text{ \AA}$ (X = I). With exception of $\text{CH}_3\text{NH}_3\text{PbCl}_3$ the compounds show intense colour, but there is no significant conductivity under normal conditions. The properties of the system are explained by a "p-resonance-bonding". The synthesis is described.

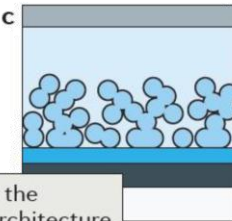


Synthesis and structural characterization of 3D HOIPs

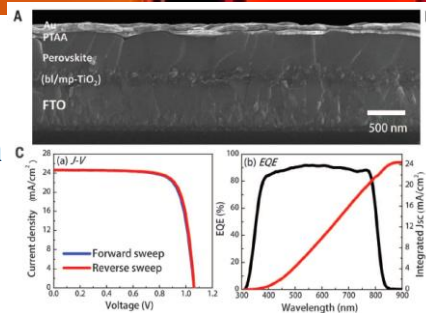
SOLAR CELLS

High-performance photovoltaic perovskite layers fabricated through intramolecular exchange

Woon Seok Yang,^{1,*} Jun Hong Noh,^{1,*} Nam Joong Jeon,¹ Young Chan Kim,¹ Seungchan Ryu,¹ Jangwon Seo,¹ Sang Il Seok^{1,2,†}



Breakthroughs in the fabrication and architecture of HOIP solar cells



1839

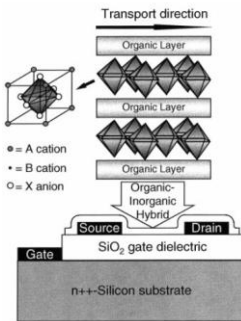
Perowskit entdeckt



1892

Organic-Inorganic Hybrid Materials as Semiconducting Channels in Thin-Film Field-Effect Transistors

C. R. Kagan, D. B. Mitzi, C. D. Dimitrakopoulos
SCIENCE VOL 286 29 OCTOBER 1999



1978

Extensive characterization and usage of HOIPs in optoelectronic devices

1994

1999

First HOIP solar cell with 4% power conversion efficiency



J|A|C|S
COMMUNICATIONS

Published on Web 04/14/2009

Organometal Halide Perovskites as Visible-Light Sensitizers for Photovoltaic Cells

Akihiro Kojima,[†] Kenjiro Teshima,[‡] Yasuo Shirai,[§] and Tsutomu Miyasaka^{*,†,||}
Graduate School of Arts and Sciences, The University of Tokyo, 3-8-1 Komaba, Meguro-ku, Tokyo 153-8902, Japan, Graduate School of Engineering, Toin University of Yokohama, and Pecell Technologies, Inc., 1614 Kurogane-cho, Aoba, Yokohama, Kanagawa 225-8502, Japan, and Graduate School of Engineering, Tokyo Polytechnic University, 1583 Iiyama, Atsugi, Kanagawa 243-0297, Japan

Received December 9, 2008; Revised Manuscript Received April 1, 2009; E-mail: miyasaka@cc.toin.ac.jp

2009

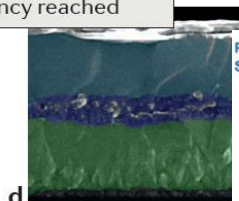
2012

2013

2015

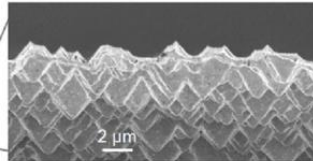
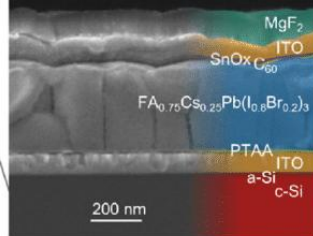
2023

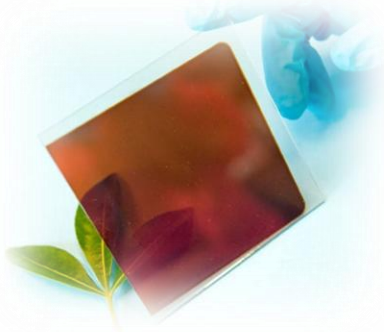
20% power conversion efficiency reached



Perovskite Solar Cell
Silicon Solar Cell

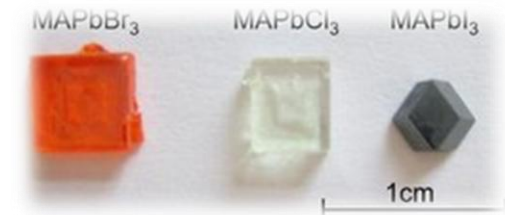
Perowskit/Silizium Tandem bricht SQ-Grenze (33.9%)





Volumenproben
(Dünnsfilme)

Einzelne (Makro)kristalle

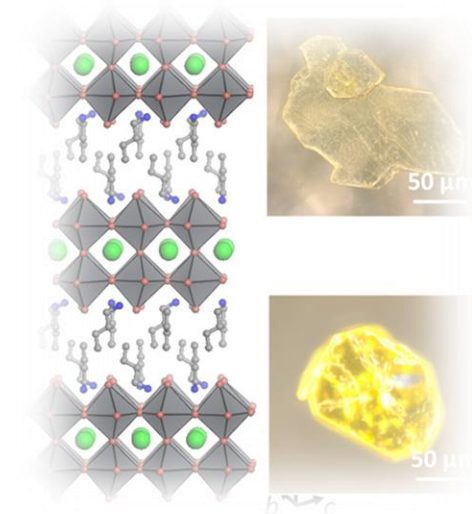


Nanokristalle

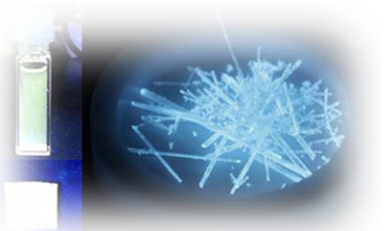
CsPbX_3 (X=Cl, Br, I) nanocrystals



Quasi-2D
Perowskite



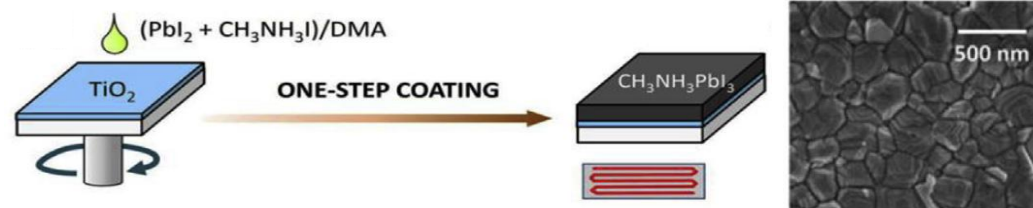
0D - Perowskite



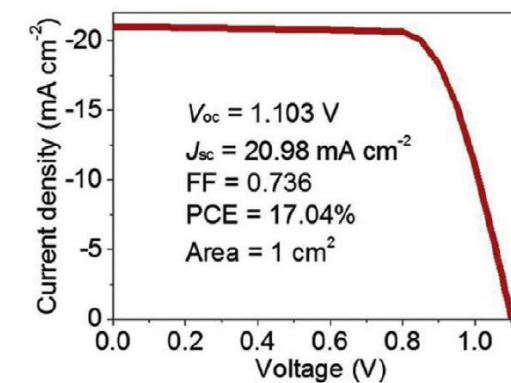
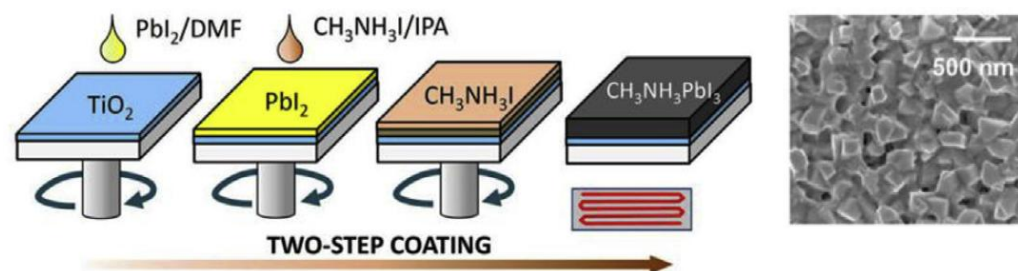
- Erste erfolgreich angewendete Methode

- Zwei Varianten:

- Einschrittverfahren



- Zweischnittverfahren



- Vorteile:

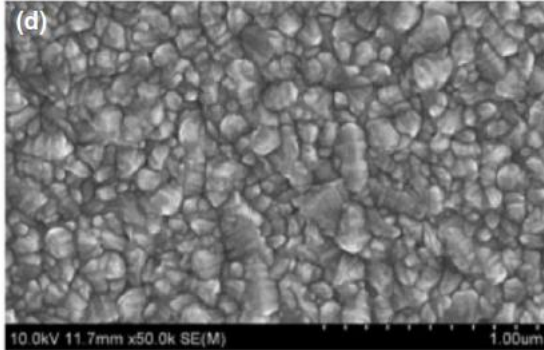
- Schnell, leicht, reproduzierbar, Dicke präzise einstellbar, hohe Qualität

- Nachteile:

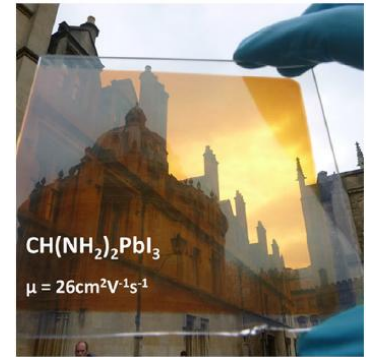
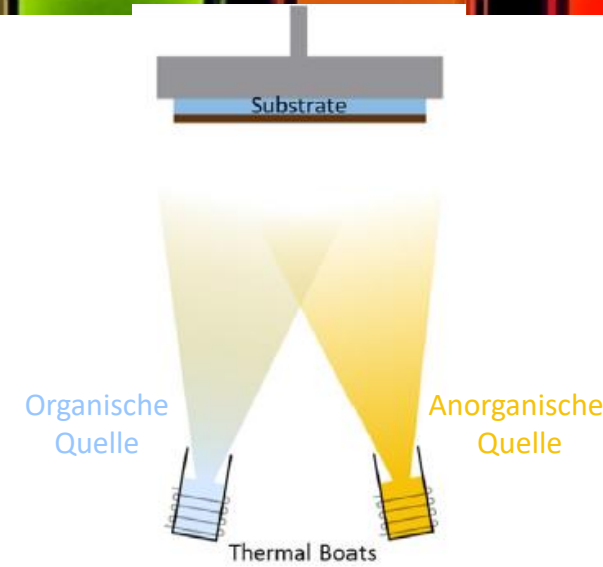
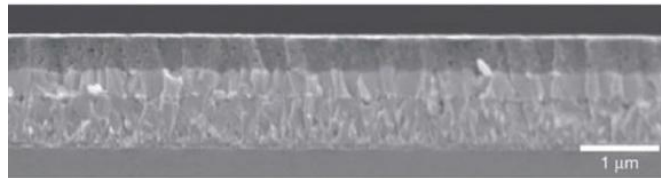
- Skalierbarkeit (Nur für wenige cm große Substrate anwendbar)

- Kurz nach dem Spin-coating entwickelt
- Substrat wird in einer Hochvakuumkammer platziert
- Organische und anorganische Quellen werden gleichzeitig auf Substrate aufgedampft

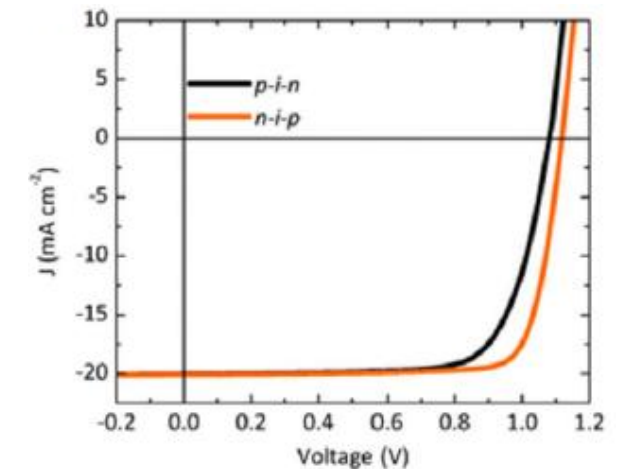
Draufsicht



Querschnitt

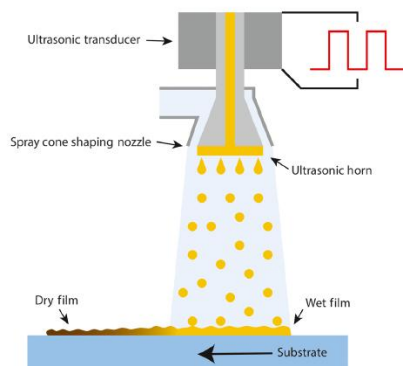


- **Vorteile:**
 - Sehr präzise Kontrolle der Dicke, sehr hohe Qualität, Reproduzierbarkeit
- **Nachteile:**
 - Herstelldauer (Vakuum), Skalierbarkeit

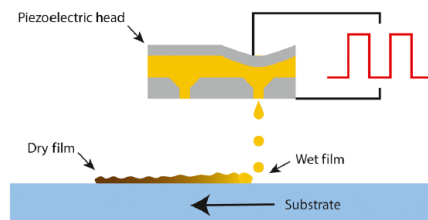


In den letzten Jahren sind immer mehr Verfahren entwickelt worden um großflächige Dünnschichten herzustellen

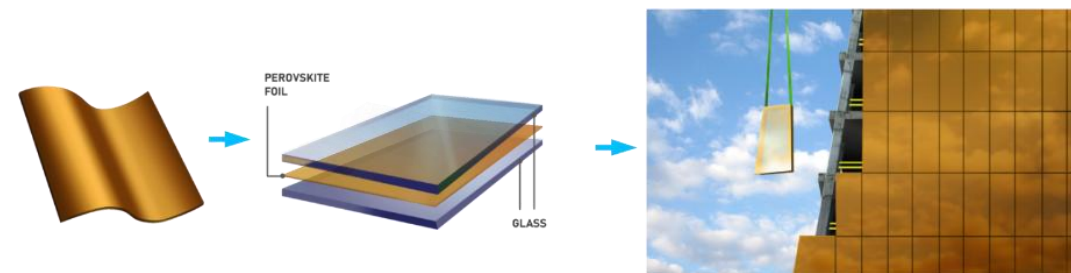
Ultraschall-Sprühverfahren



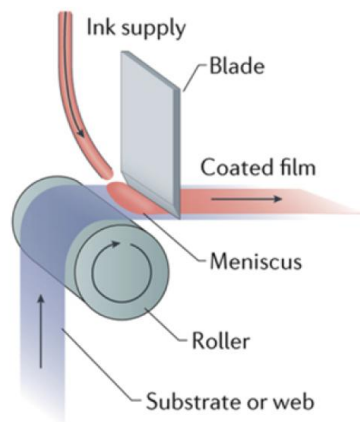
Tintenstrahldruck



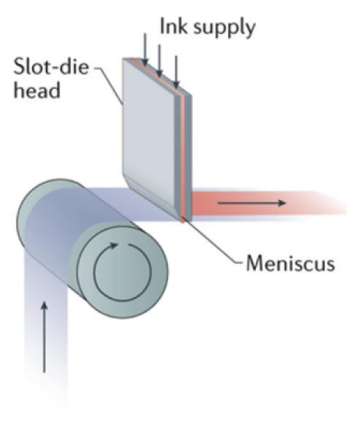
Photovoltaisches Perowskitglas



Blade coating



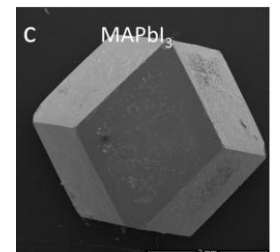
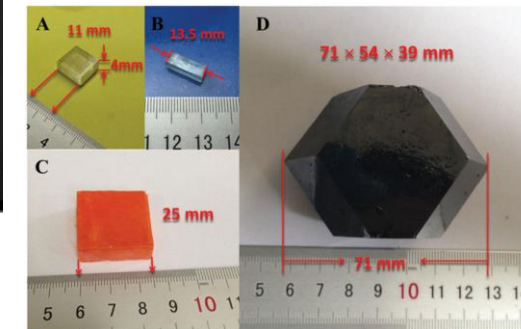
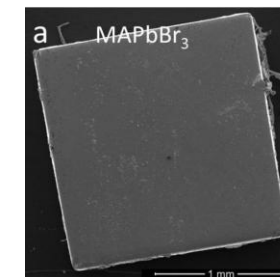
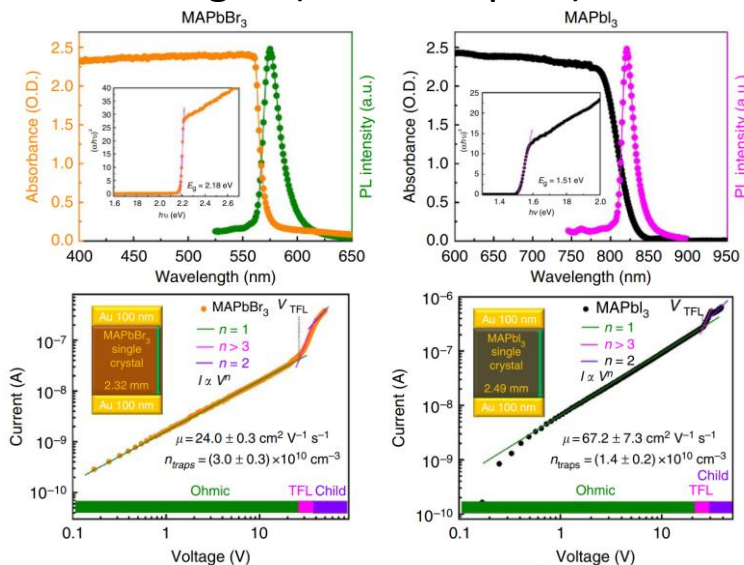
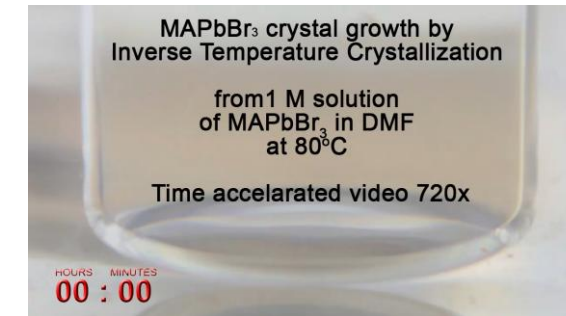
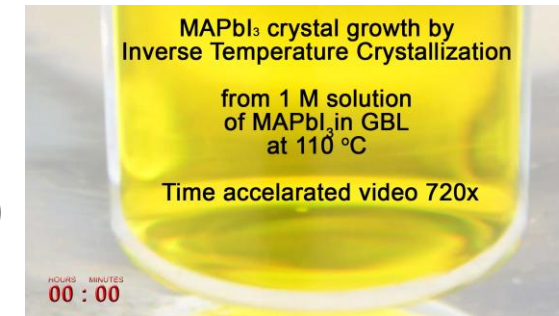
Slot-die coating



Photovoltaische Jalousien



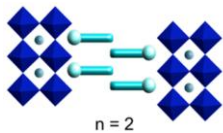
- Alle Dünnschichten sind polykristallin (Defekte, Korngrenzen)
→ Herstellung von makroskopischen Einzelkristallen
- **Inverse Temperatur-Kristallisation (ITC)**
 - Perowskitedukte sind bei niedrigen Temperaturen besser löslich
 - Erst: Gesättigte Lösung bei niedriger Temperatur herstellen
 - Danach: Lösung erhitzen
 - Ergebnis: cm große Kristalle (nach mehreren Wiederholungen)
 - Vereinfachte Messungen (z.B. Transport)



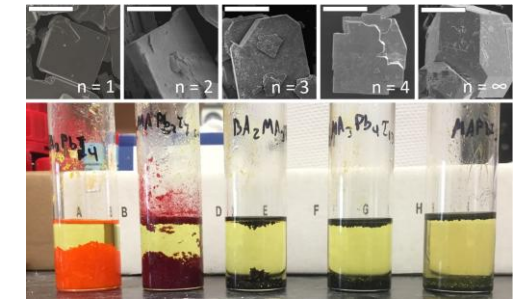
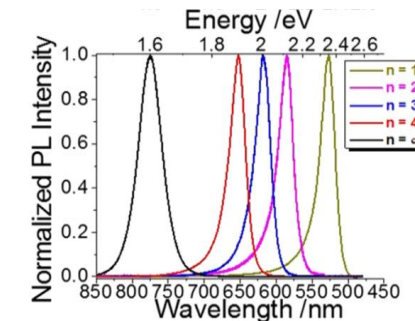
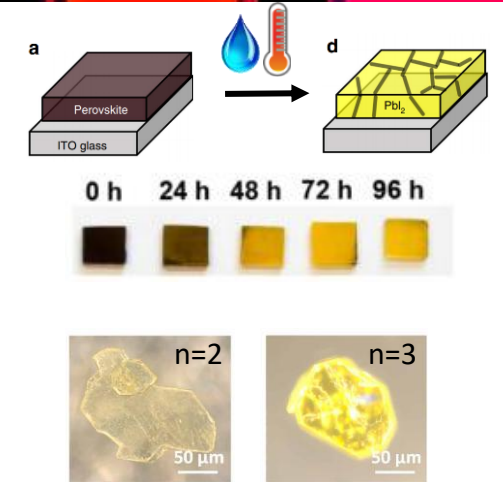
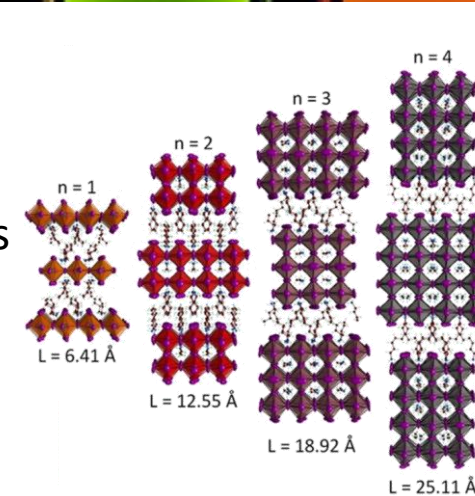
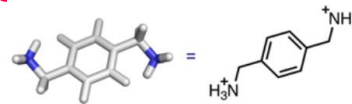
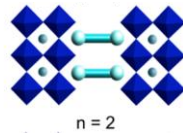
- 3D Perowskite sind nicht besonders stabil, degradieren schnell
- Lange, hydrophobe Liganden schützen den Perowskit (Ummantelung)
- Wenn die Ligandenkonzentration hoch ist, passiert etwas Unerwartetes
 - Es bilden sich zweidimensionale Perowskite
 - Die Dicke der Perowskitschicht wird über die Ligandenkonzentration gesteuert
 - Das hat eine starke Auswirkung auf die optischen Eigenschaften
- Anordnung der Schichten über die Liganden gesteuert

→ Ruddlesder

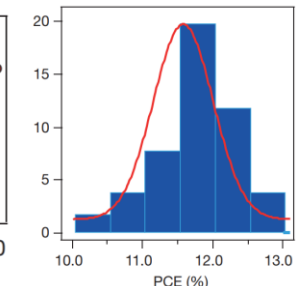
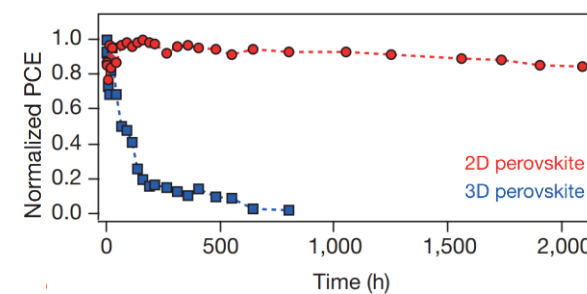
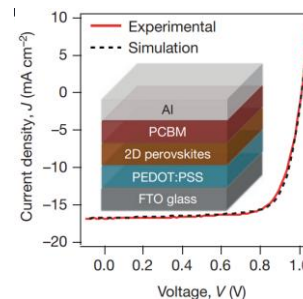
Ruddlesden-Popper



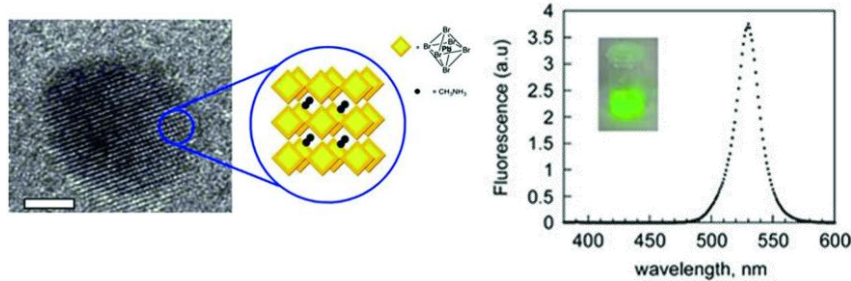
Dion-Jacobson

Phasenⁿ

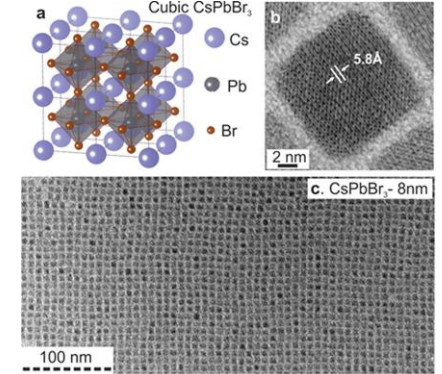
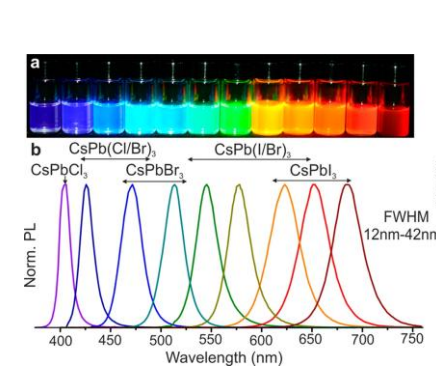
- Anwendung: Optoelektronik (z.B. Solarzellen)
 - Stabilität stark erhöht
 - Ladungsdichte und damit PCE niedrig



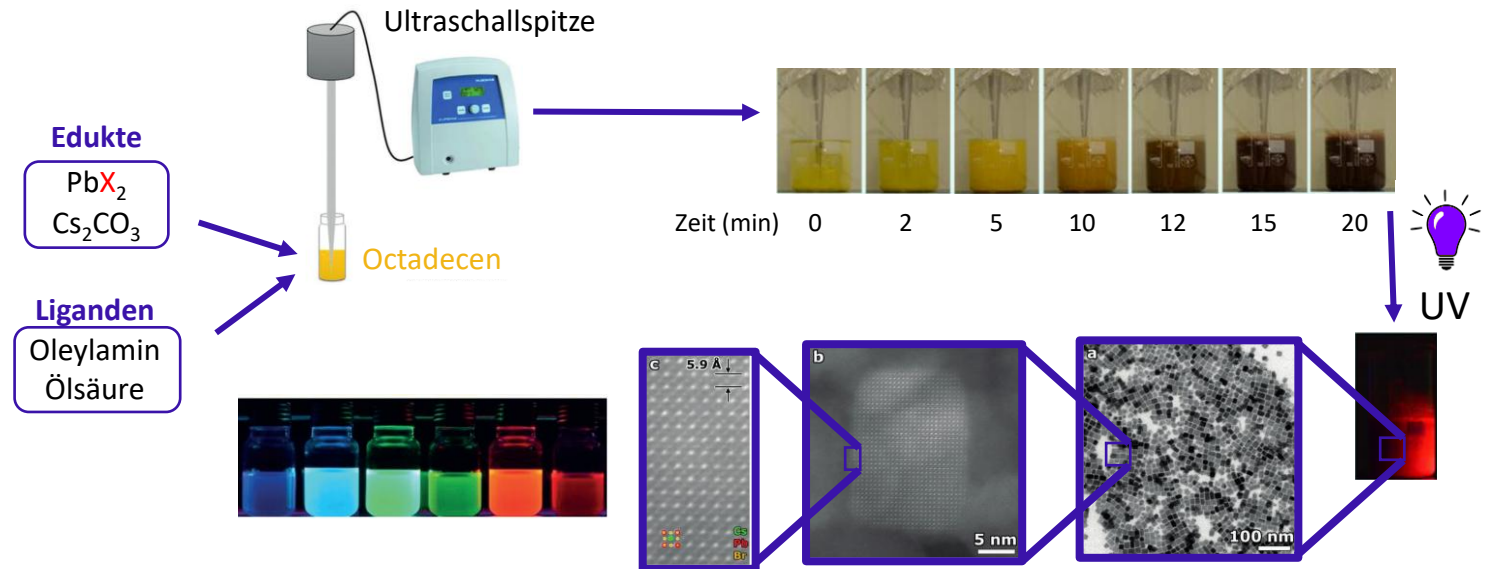
- Erste Herstellung 2014



- Herstellverfahren wurden aus der gewöhnlichen QD-Synthese schnell übernommen (Heißeinjektion)

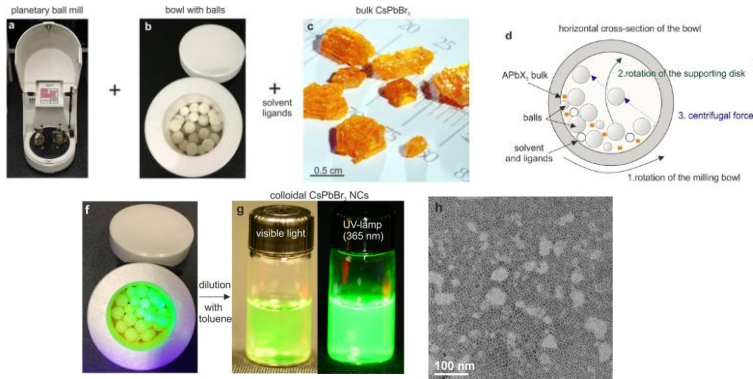


- Eine Niedrigtemperatursynthese funktioniert auch:

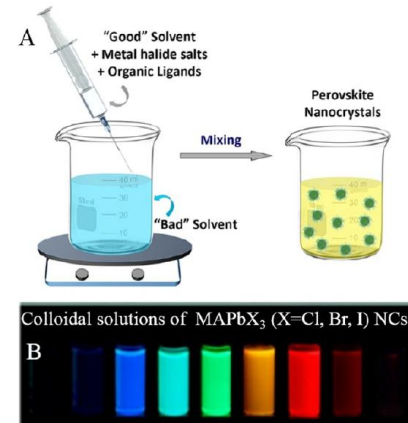


- Andere Niedrigtemperaturverfahren können auch verwendet werden, um Perowskite zu erzeugen:

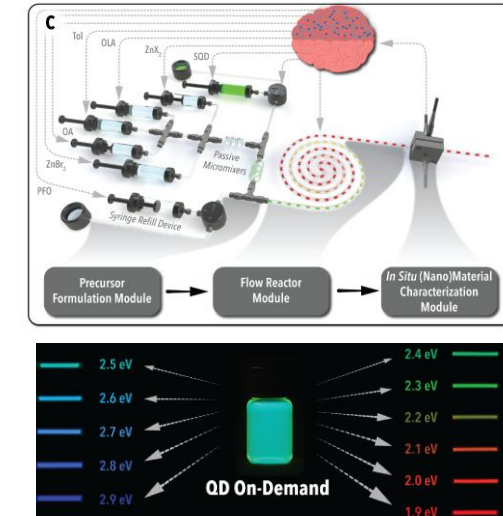
Kugelmahlen



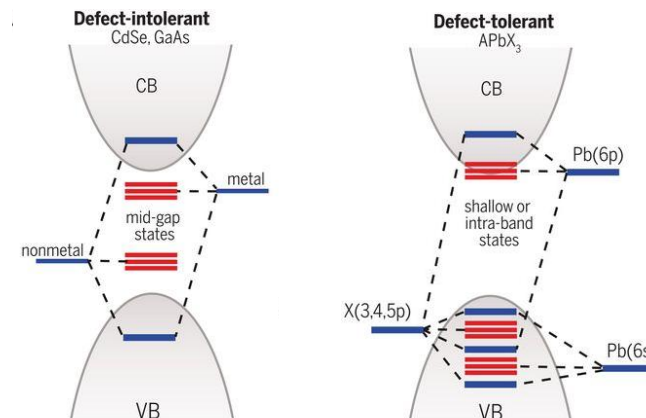
Ligand-unterstütztes Umfällen



Automatisierte Synthesen

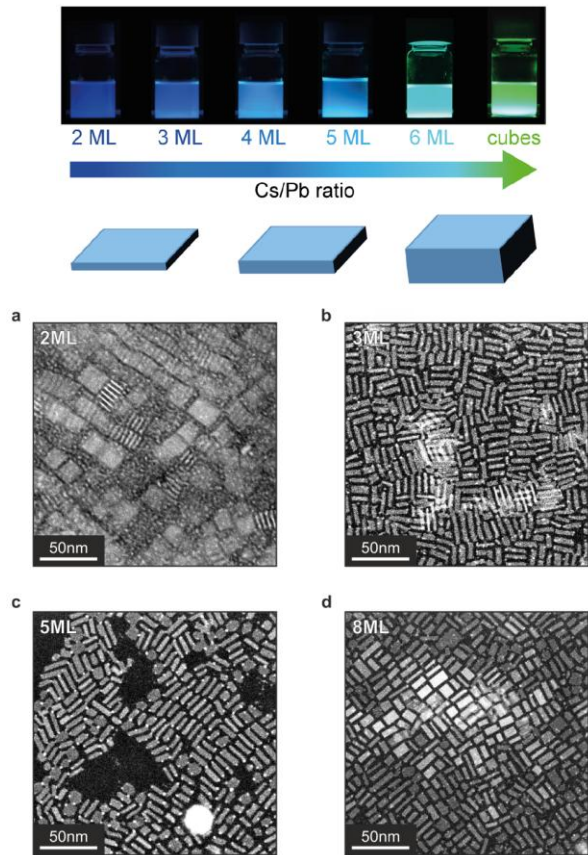


- Gewöhnliche QD benötigen hoch kontrollierte Synthesebedingungen
Warum ist die Perowskitesynthese so einfach?
 - Halidperowskite sind defekttolerant!
 - Passivierung ist nicht so kritisch

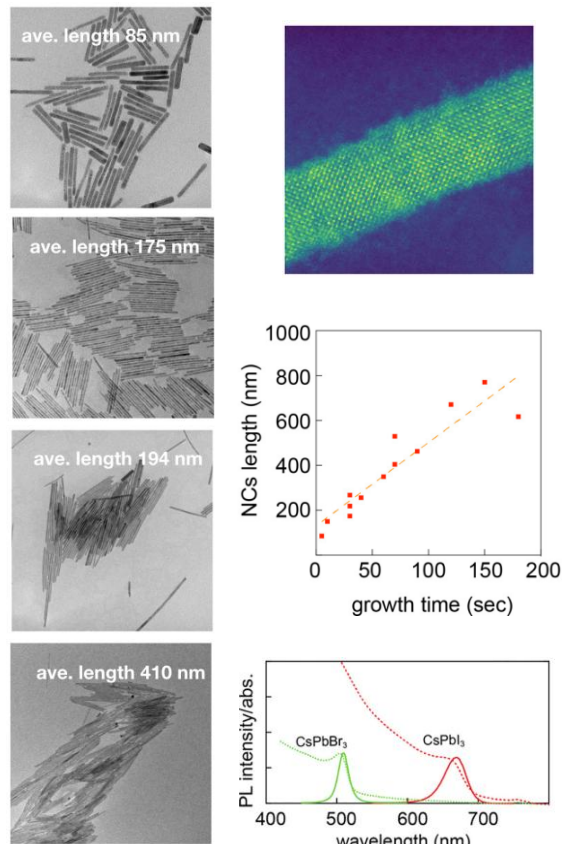


- Präzise Kontrolle von Komposition, Morphologie und Zusammensetzung

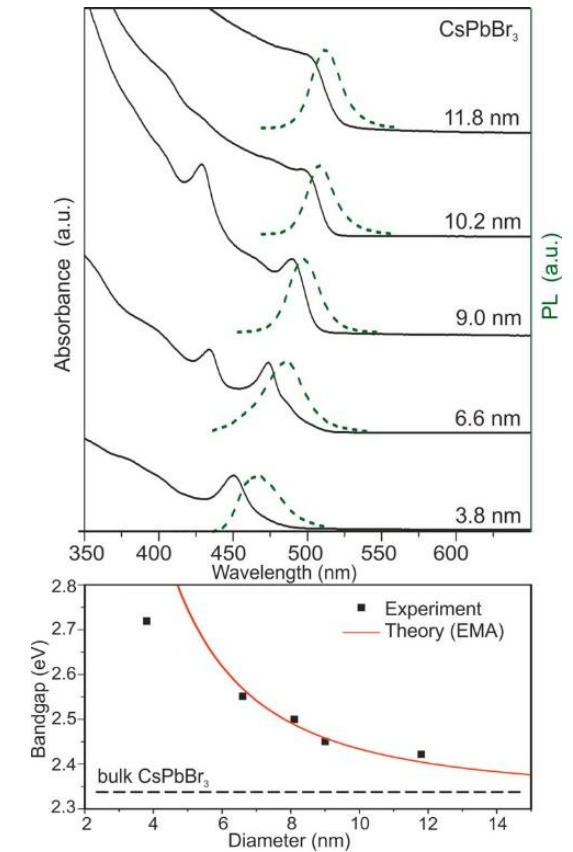
2D



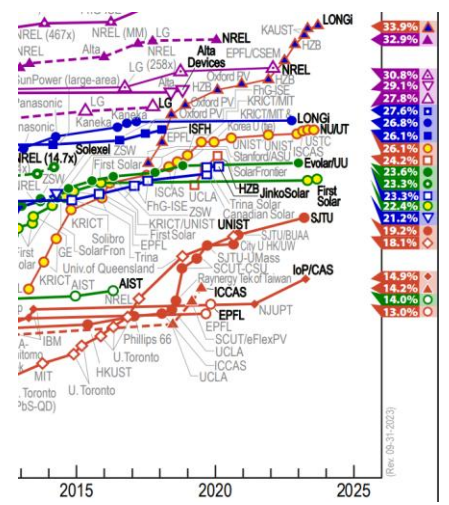
1D



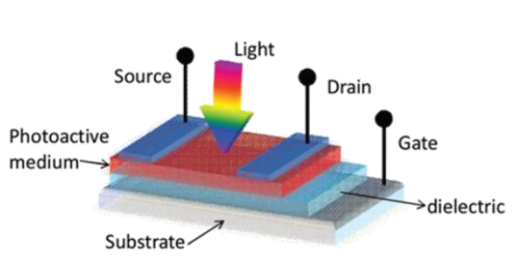
0D



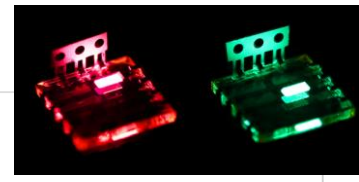
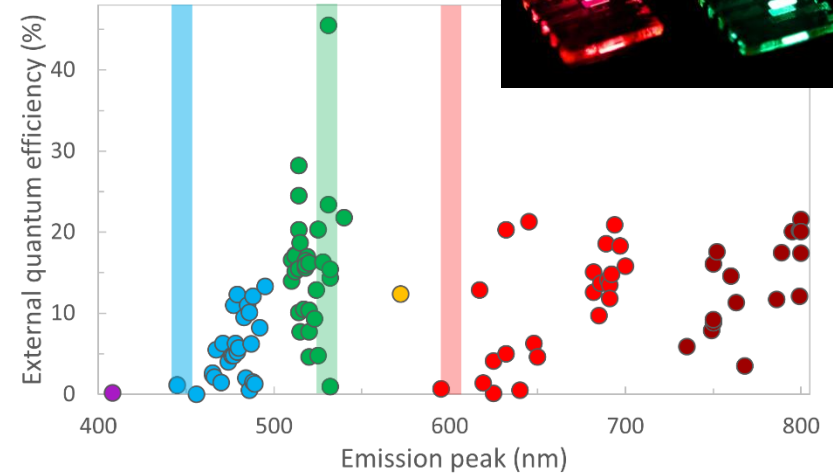
Solarzellen



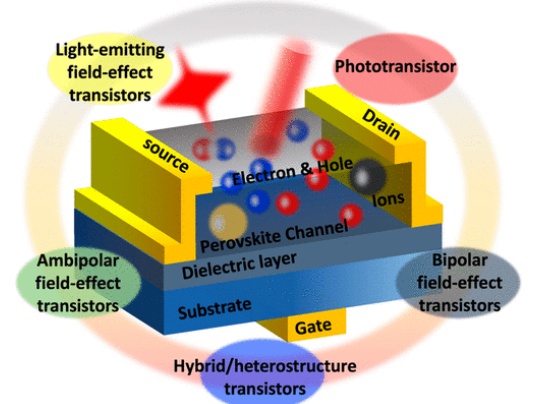
Photodetektoren



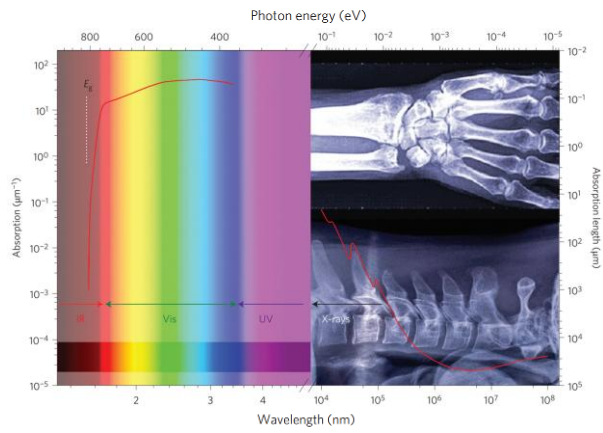
LEDs



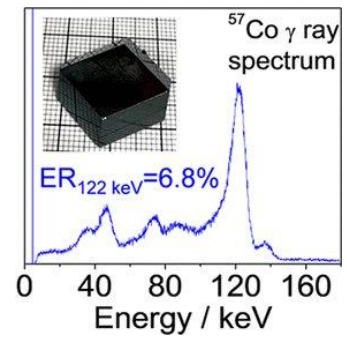
Transistoren



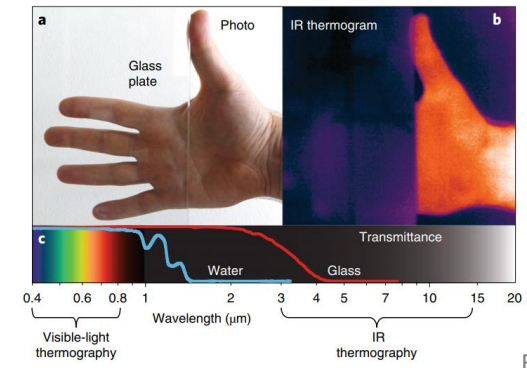
Röntgendetektor



Gammastahlungdetektor



Wärmebildkamera

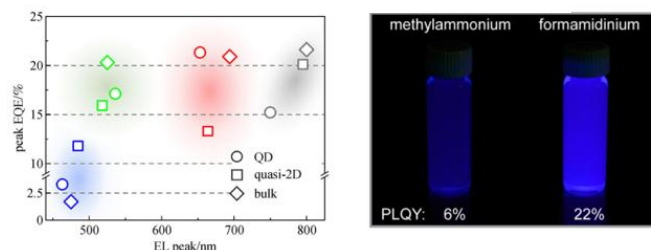


F. Li et al. *Acc. Mater. Res.* 3, 8-20 (2022)
 S. Yakunin et al. *Nat. Mater.* 18, 846-852 (2019)
 S. Yakunin et al. *Nat. Photon.* 10, 585-589 (2016)
 S. Yakunin et al. *NPG Asia Mater.* 9, e431 (2017)

J. Yuan et al. *Joule* 4, 1160-1185 (2020)
 J. Park et al. *Trends Chem.* 2, 837-849 (2020)
 Y. Hu et al. *ACS Photon.* 4132-4138 (2018)
 J. Miao and F. Zhang *J. Mater. Chem.* 7, 1741-1791 (2019)

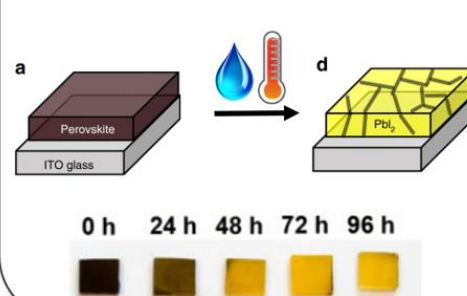
- Trotz der außergewöhnlichen Entwicklung, gibt es Probleme, die (noch) die Kommerzialisierung limitieren

Spektrale Effizienz

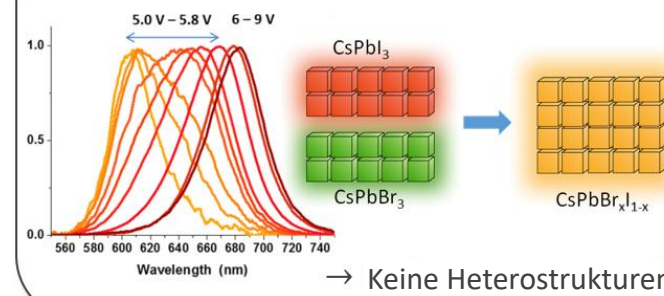


→ Blaue Perowskite sind viel ineffizienter

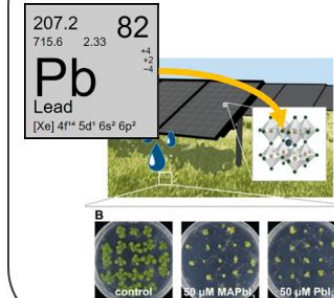
Stabilität



(Halid)ionen-Diffusion

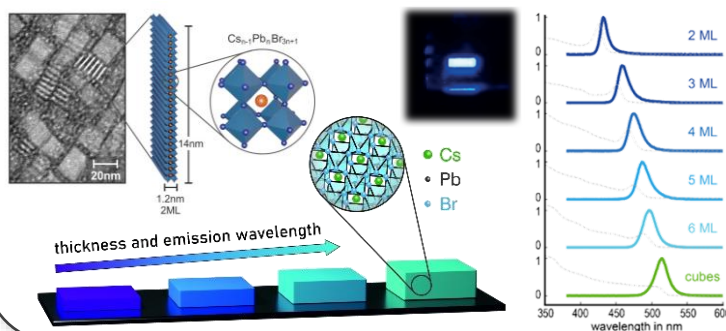


Toxizität

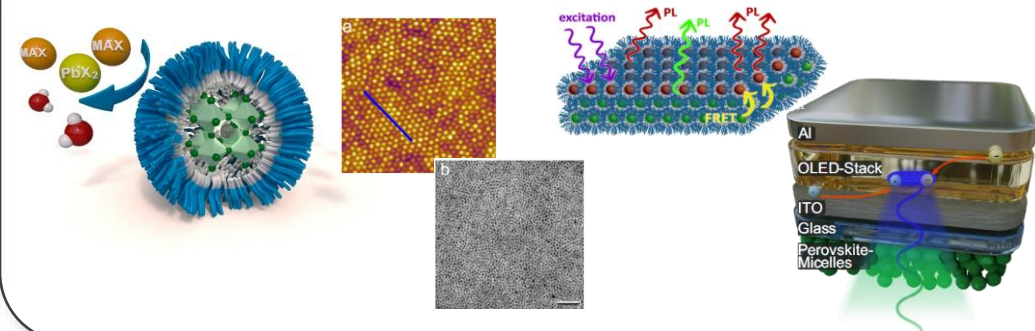


Um diese Probleme anzugehen, haben wir unterschiedliche Nanokristallsysteme entwickelt

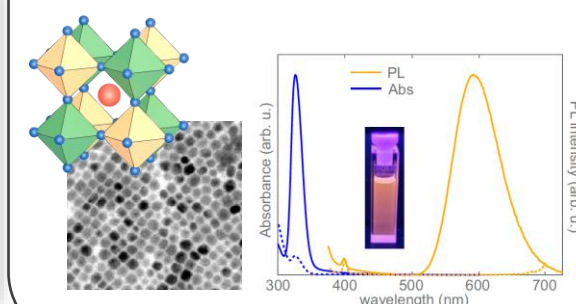
2D Nanoplättchen



Polymermizellenhüllen



Doppelperowskite

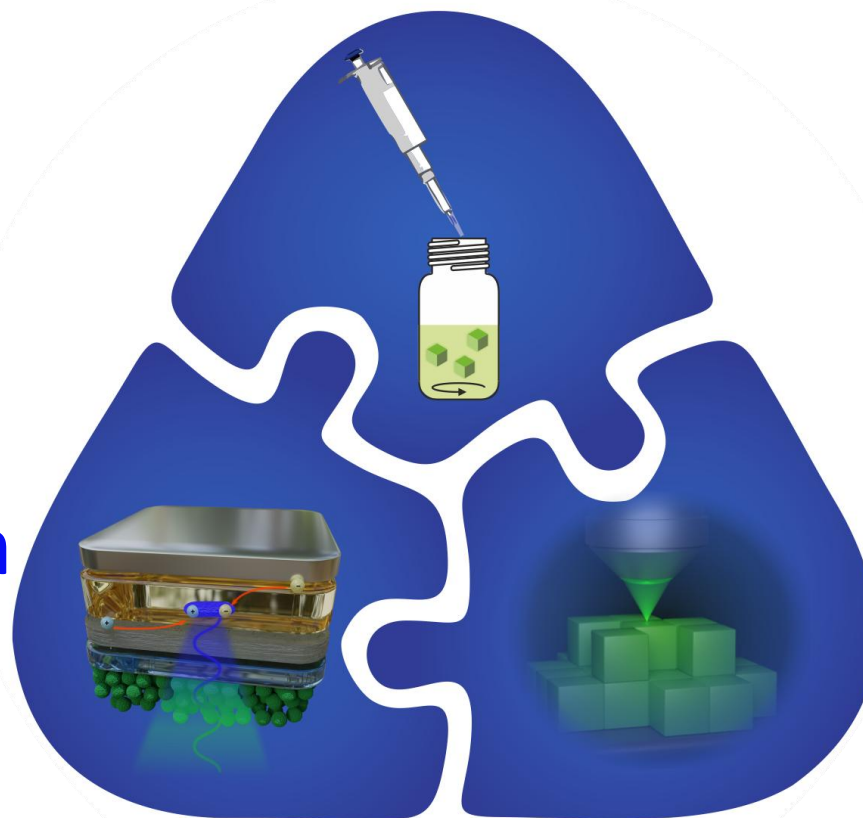


Synthese

Neue (Nano)materialien entdecken

Anwendungen

Perowskite für ihren Einsatz
in der Optoelektronik und
Energieumwandlung
optimieren

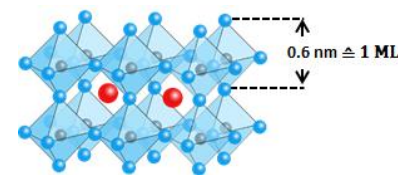


Spektroskopie/ Mikroskopie

Die ihnen zugrunde
liegenden Eigenschaften
verstehen

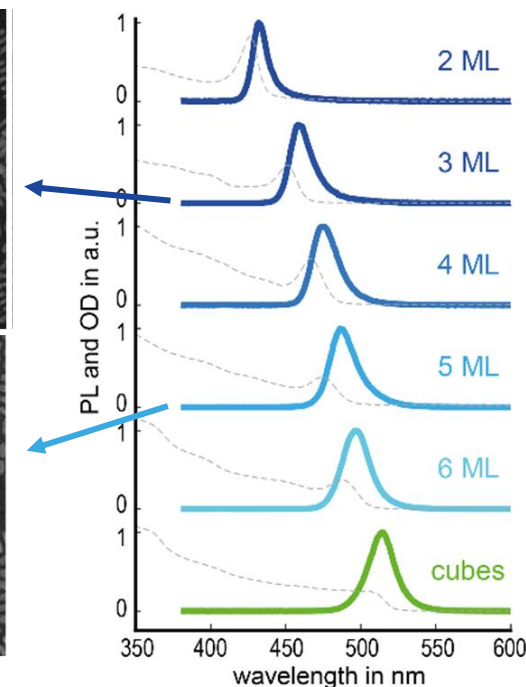
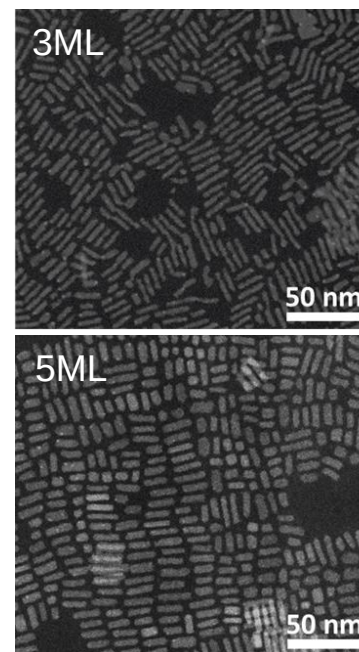
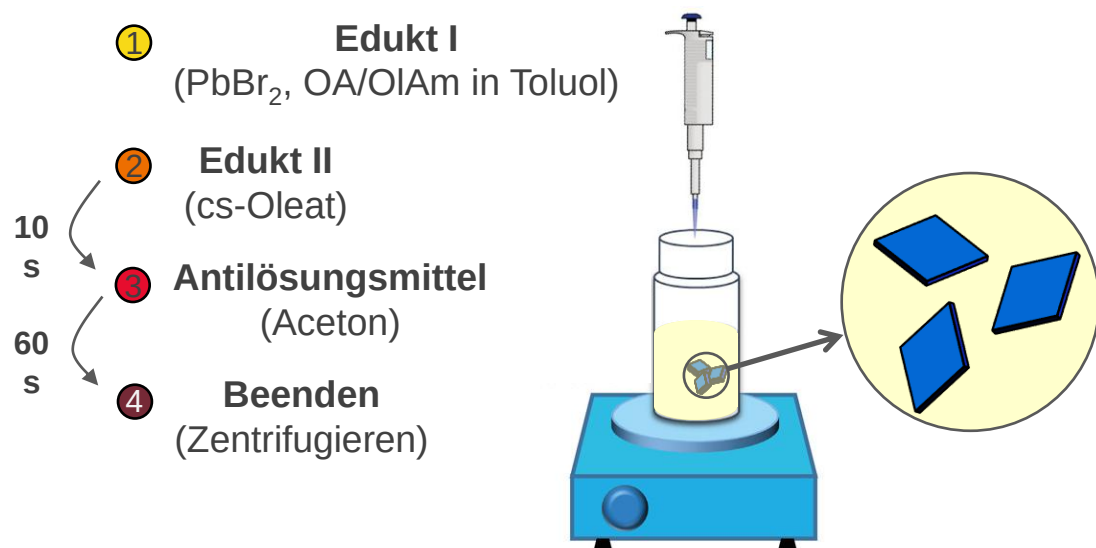


- Synthese wird bei Raumtemperatur und an Luft durchgeführt
- Erfolgt ähnlich der Ligand-unterstützten Umfällung



Monolagen-präzise Einstellbarkeit der Dicke

QY



Pb/Cs Eduktverhältnis

7

9

15

28

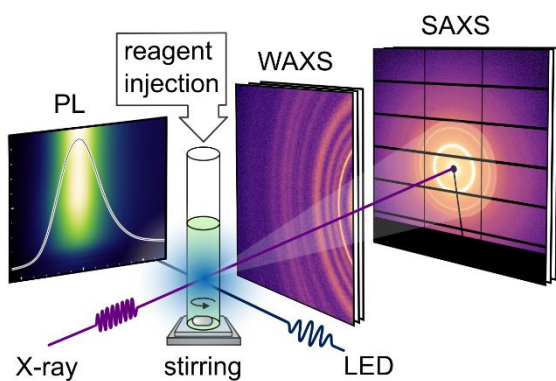
42

95

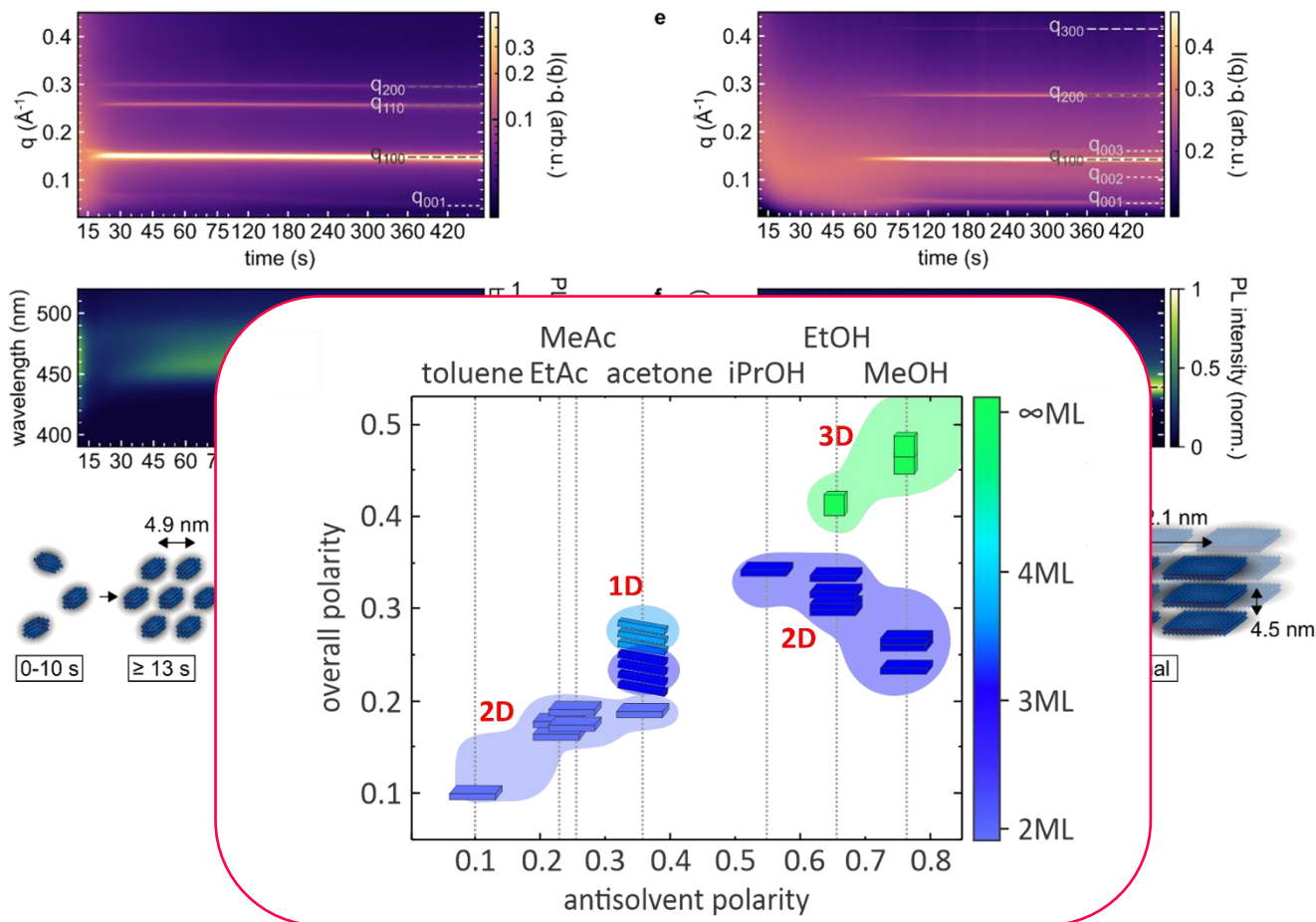
Wie läuft die Synthese ab?



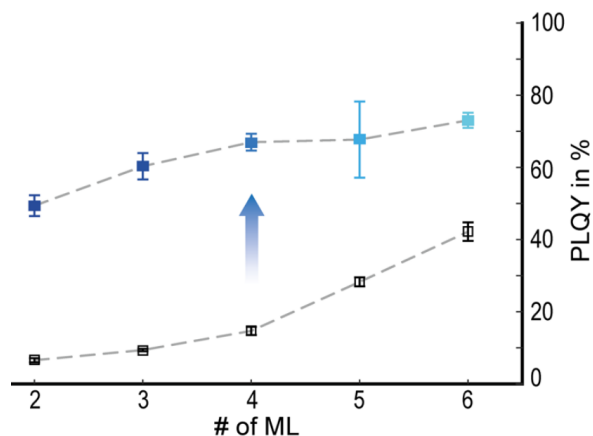
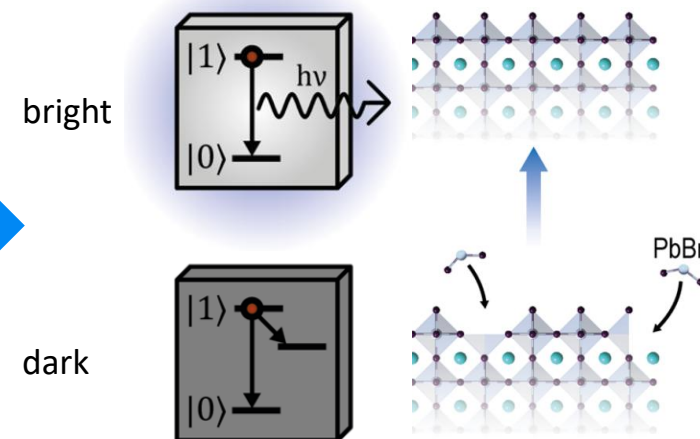
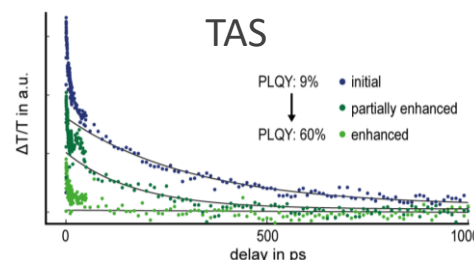
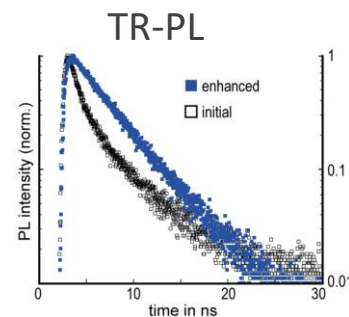
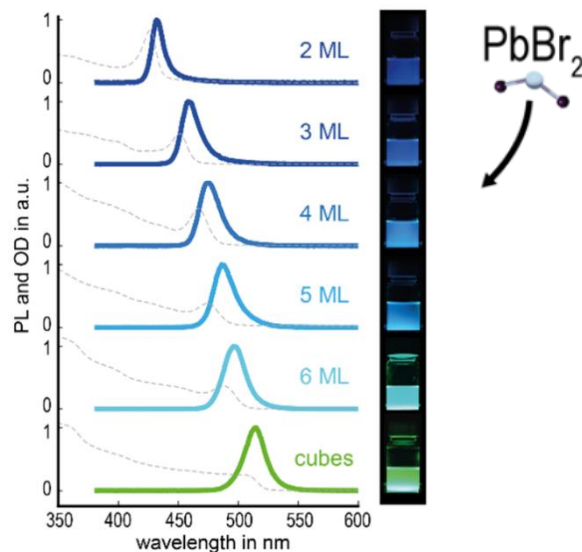
- Um die Synthese weiter zu verbessern, muss man verstehen, was vor sich geht
- Optische Daten alleine sind nicht aussagekräftig genug
- PL-Spektroskopie mit Röntgenbeugung koppeln
 - ⇒ Synthesekammer entwickelt für den Einsatz am Synchrotron (DESY)
 - ⇒ Blick auf die unterschiedlichen Schritte



- ⇒ Wir verstehen, warum manchmal 2D und manchmal 1D Strukturen entstehen
- ⇒ Mit dem Wissen können wir die Morphologie kontrollieren



Quanteneffizienz erhöhen durch Oberflächenpassivierung



- Quanteneffizienz der Nanoplättchen direkt nach der Synthese sehr niedrig
- Nachträgliches Addieren einer PbBr_2 -Ligand-Lösung erhöht die Quanteneffizienz massiv
- Aus transientser Absorptions- und zeitaufgelöster PL-Spektroskopie ermitteln wir:
- Die zusätzliche Lösung passiviert Oberflächendefekte und macht dunkle Nanoplättchen hell

Kann maschinelles Lernen uns bei der Entwicklung neuer Materialien weiterhelfen

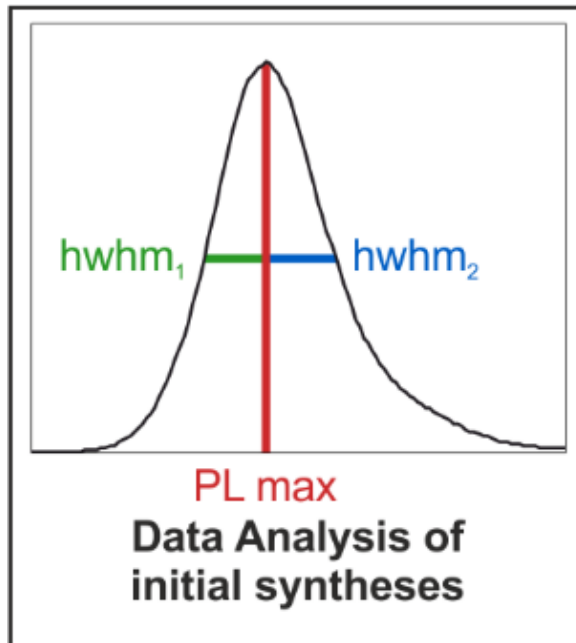


Bis jetzt: Erfolg durch systematisches Ausprobieren

Neuer Ansatz: Maschinelles Lernen verwenden

Ansatz: 30
Synthesen

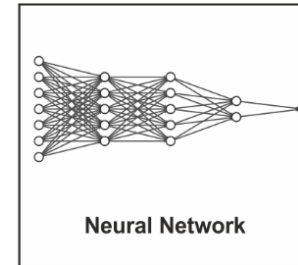
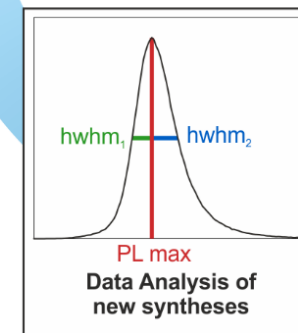
Ausleseparameter:
Emissionsqualität (hwhm)
und spektrale Position



Eingangsgrößen:
Eduktmengen (3x)



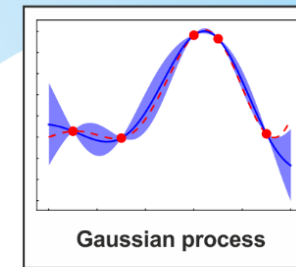
15 zusätzliche
Synthesen pro
Durchlauf



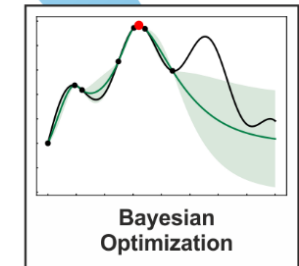
Neural Network



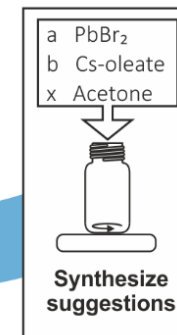
Random Forest

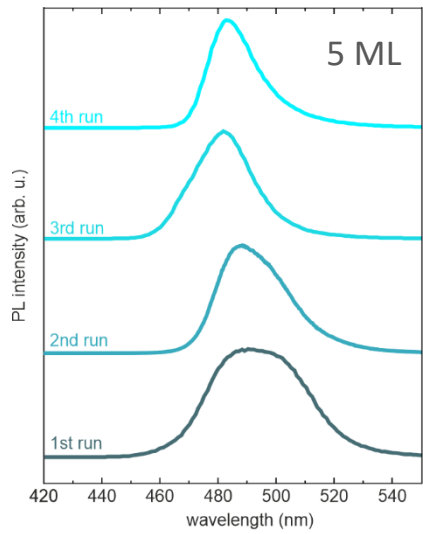
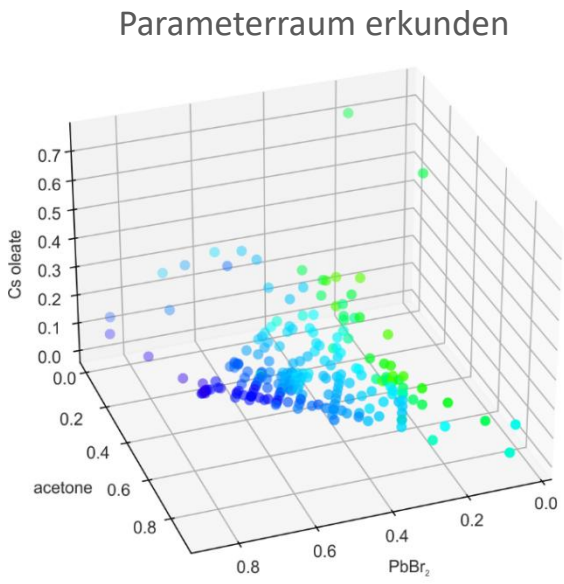


Gaussian process



Bayesian
Optimization

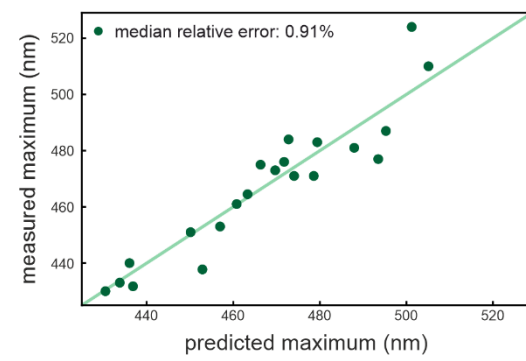




Optische Eigenschaften werden stets besser

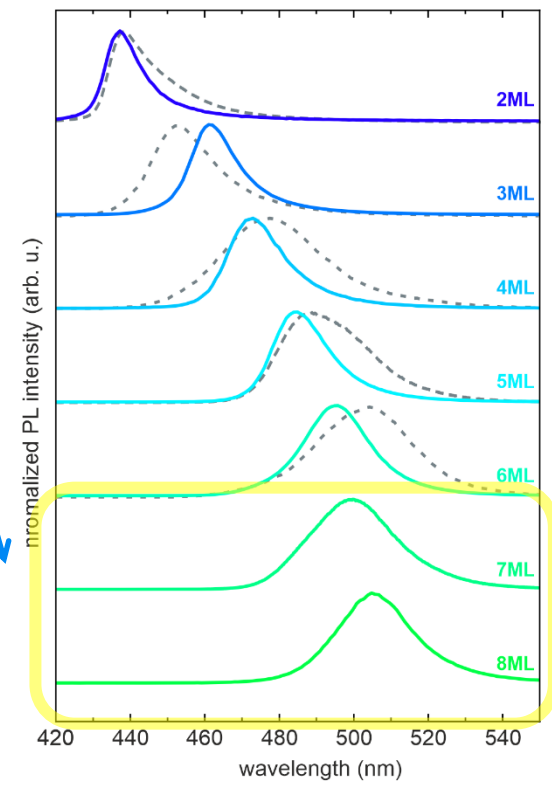


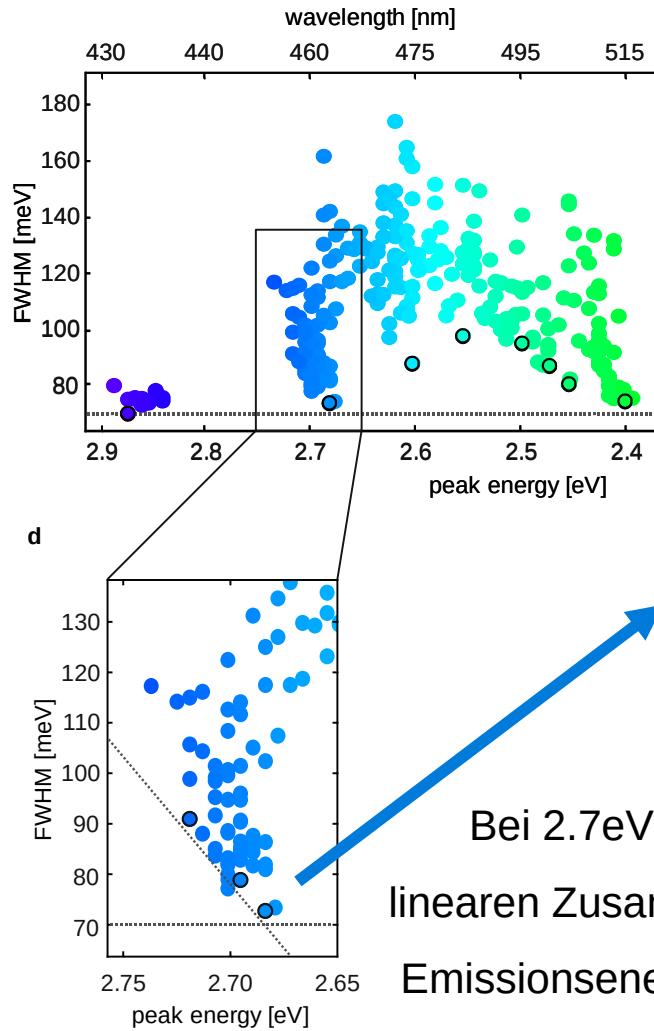
Quantenausbeuten stark erhöht (bis zu 95%)



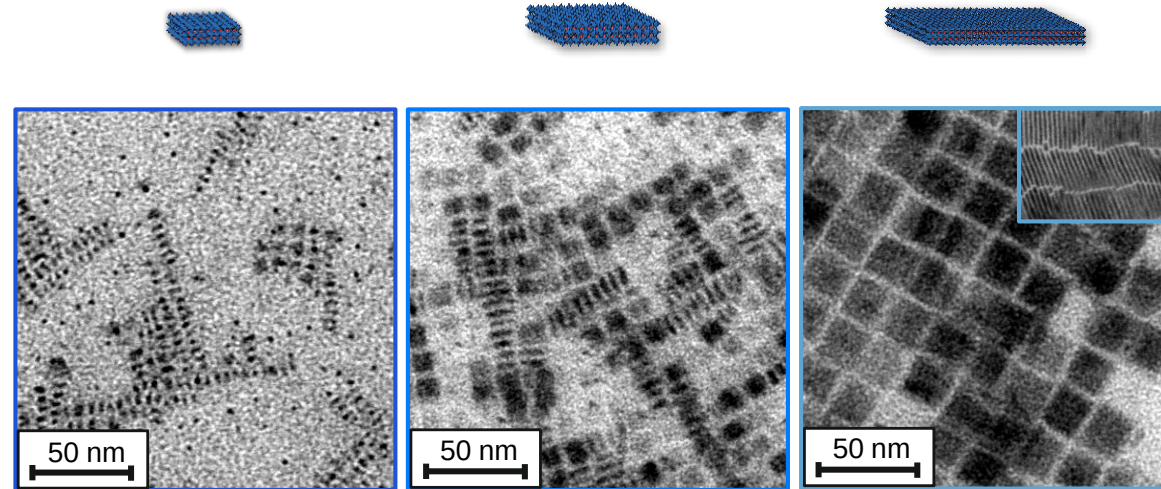
Algorithmus kann auch das PL-Emissionsmaximum anhand der Eduktparameter voraussagen

All Dicken verbessert
Neue, dickere NPL erhalten!



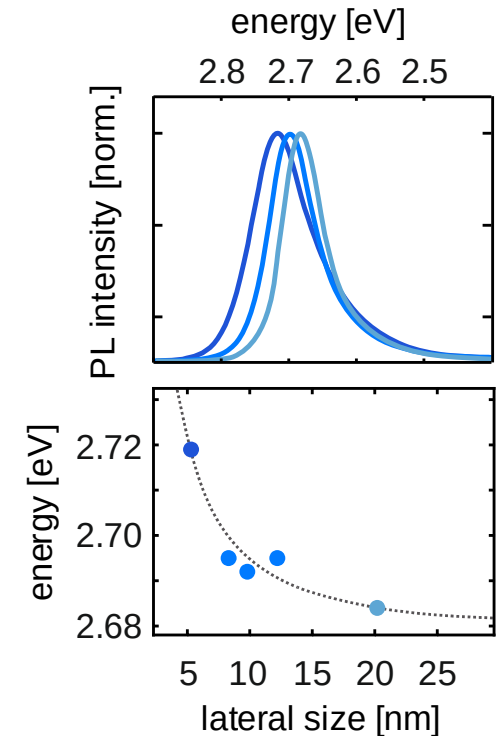


In TEM sieht man 2D Nanoplättchen
mit zunehmender Seitenlänge



Bei 2.7eV sieht man einen
linearen Zusammenhang zwischen
Emissionsenergie und Peakbreite

Durch PL Spektroskopie
bestätigt

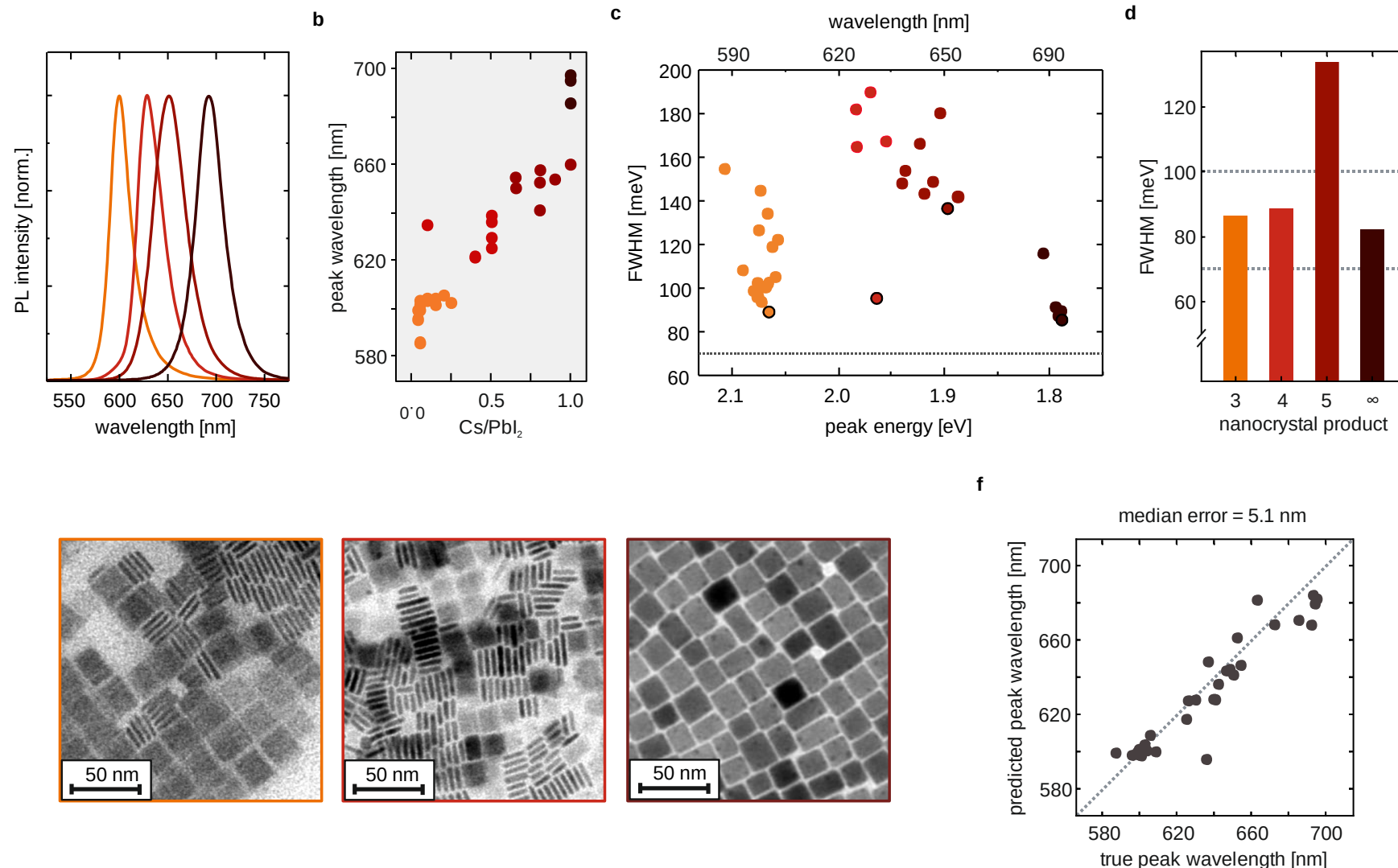


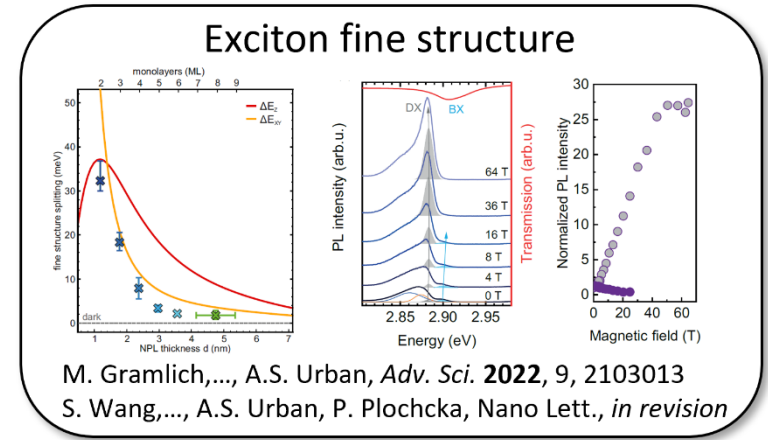
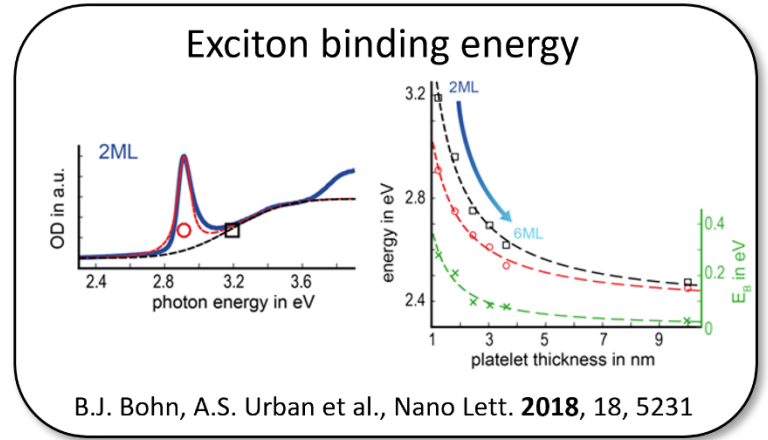
Nun können wir also nicht nur die Dicke sondern auch die laterale

Größe der 2D Nanokristalle einstellen

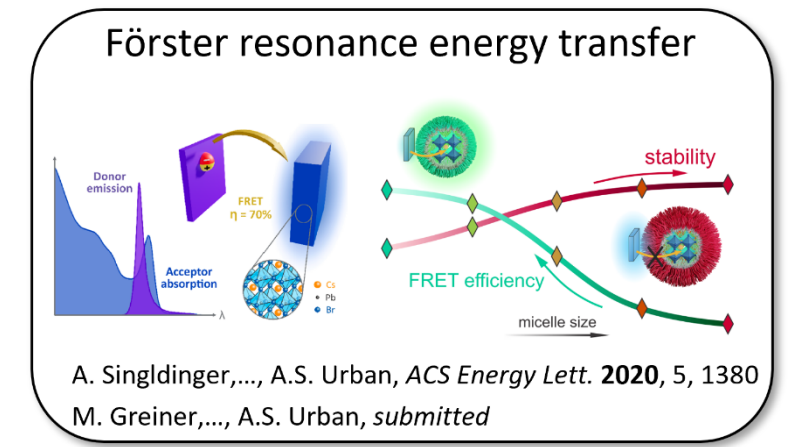
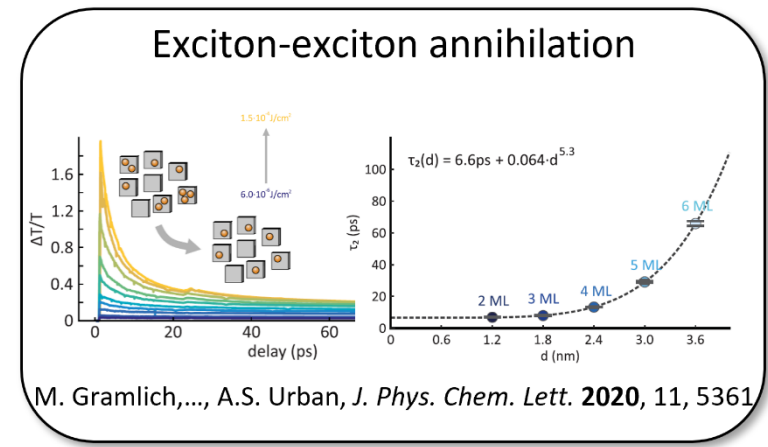
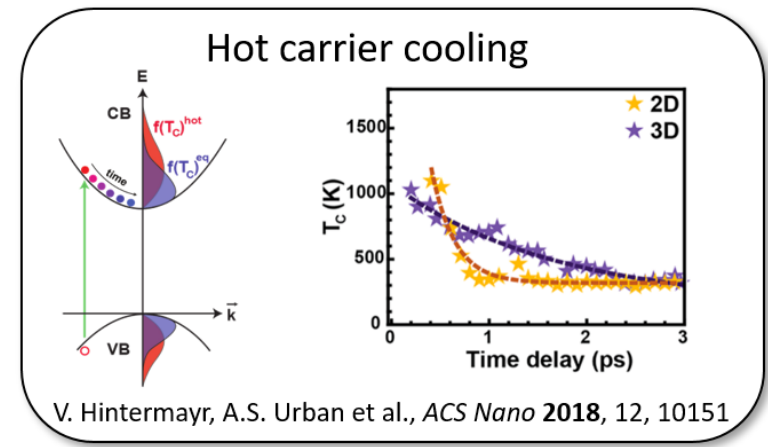
Mit unserem neuen
Algorithmus können wir auch
schnell andere Materialien
verbessern

Hier: CsPbI_3



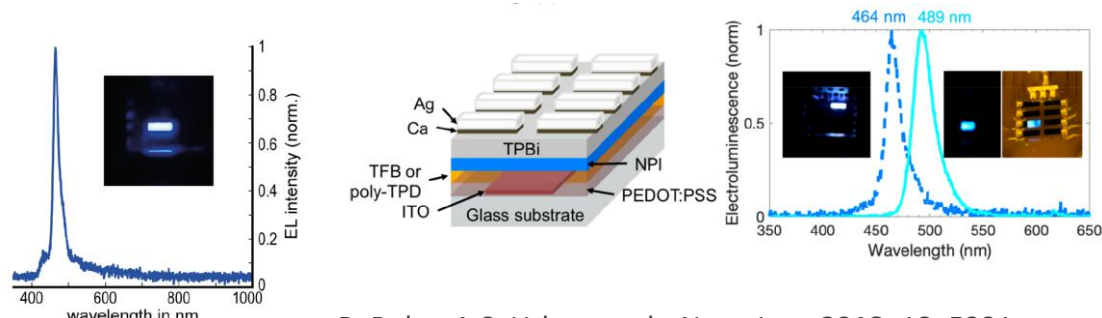


Die sehr präzise Dickenkontrolle erlaubt eine Untersuchung von exzitonischen Effekten



- Um die Qualität der neuen Materialien zu testen, fertigen wir Testbauteile an
- Hauptfokus: Lichterzeugung und Energieumwandlung

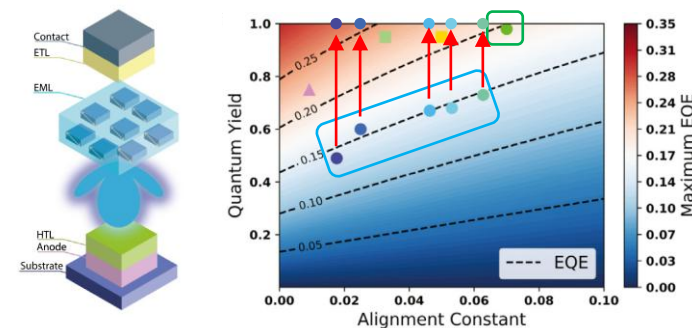
2D NPL für blaue LEDs



B. Bohn, A.S. Urban et al., *Nano Lett.* **2018**, 18, 5231

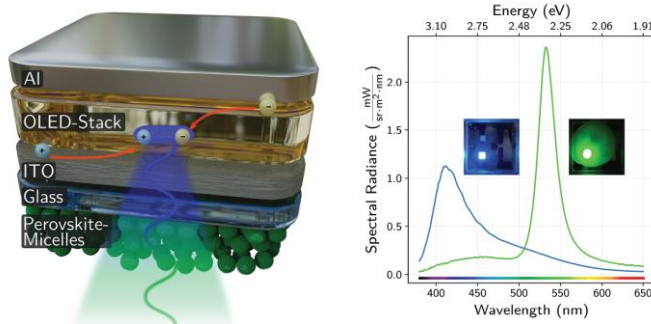
R.L. Hoyer, A.S. Urban et al., *ACS Energy Lett.* **2019**, 4, 1181

Anisotropie der PL Emission von NPL erhöht die Auskopplung enhances outcoupling → höhere EQE



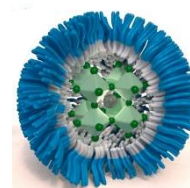
T. Morgenstern, A.S. Urban et al., *J. Lumin.* **2020**, 220, 116939

Lumineszente Farbwandler

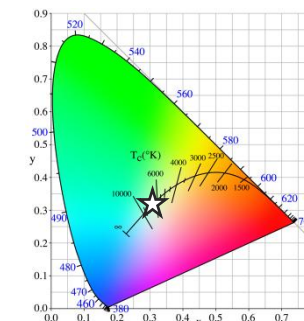


Q. Xue, ..., A.S. Urban, *Adv. Opt. Mater.* **2022**, 2102791

Mizellenperowskite



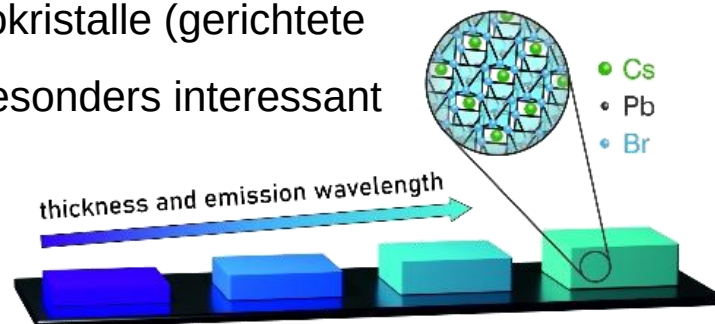
Weisslicht-LEDs



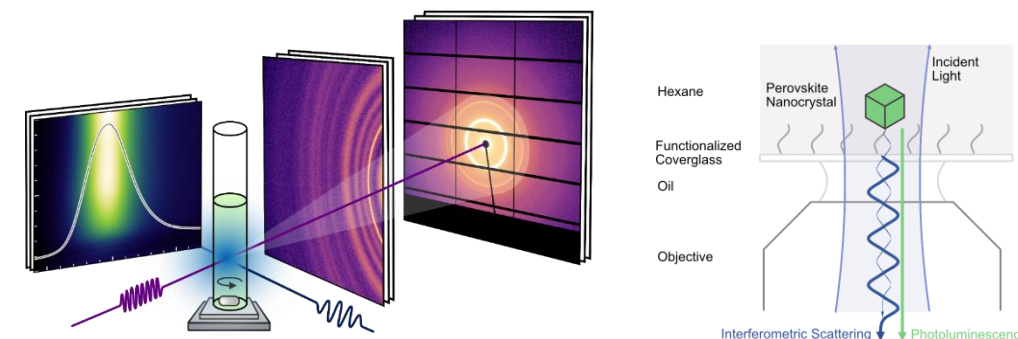
Q. Xue, ..., A.S. Urban, *unpublished*



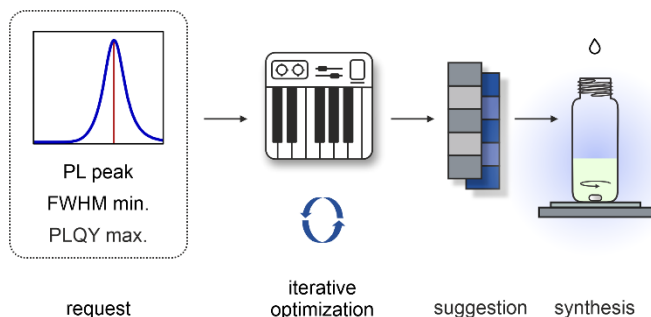
Anisotrope Nanokristalle (gerichtete Emission) sind besonders interessant



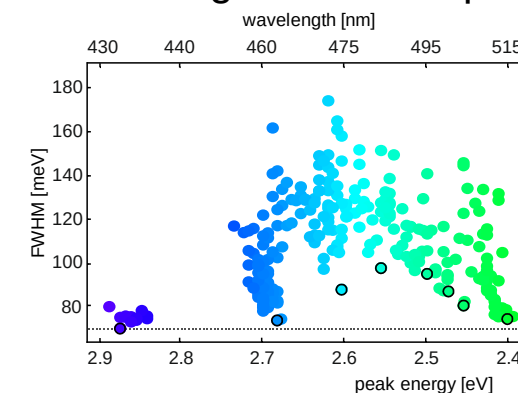
Halidperowskitnanokristalle sind die ideale Basis für die nächste Generation der Optoelektronik



Verständnis der Synthesemechanismen sind wichtig um sie zu optimieren



Machine learning kann Materialentdeckung und Optimierung enorm beschleunigen





Kollaborationen



Bert Nickel
Emiliano Cortés
Knut Müller-
Caspary



Alessio Gagliardi



Wolfgang Brütting



- 34

NANO
SPECTROSCOPY

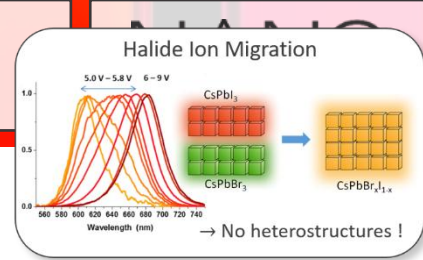
-
- Figure 1 is a scatter plot showing the band gap (eV) of various perovskite materials. The y-axis represents the band gap in eV, ranging from 0.0 to 2.8. The x-axis lists the materials, color-coded by their base cation: Pb-based (black), Sn-based (blue), Ge-based (red), Sb-based (olive), and Bi-based (teal). The materials are arranged in order of increasing band gap from left to right. The legend indicates the following categories:
- Pb-based (black)
 - Sn-based (blue)
 - Ge-based (red)
 - Sb-based (olive)
 - Bi-based (teal)

- | | | | | | | | | | | | | | | | | | | |
|--|---|--|--|---|---|--|---|---|---|--|--|---|--|--|--|---|---|-------------------------------------|
| 1
H
hydrogen
(gas) | | | | | | | | | | | | | | | | | 2
He
helium
(gas) | |
| 3
Li
lithium
(solid) | 4
Be
beryllium
(solid) | | | | | | | | | | | | | | | | | 10
Ne
neon
(gas) |
| 11
Na
sodium
(solid) | 12
Mg
magnesium
(solid) | 13
Al
aluminum
(solid) | 14
Si
silicon
(solid) | 15
P
phosphorus
(solid) | 16
S
sulfur
(solid) | 17
Cl
chlorine
(gas) | 18
Ar
argon
(gas) | | | | | | | | | | | 36
Kr
krypton
(gas) |
| 19
K
potassium
(solid) | 20
Ca
calcium
(solid) | 21
Sc
scandium
(solid) | 22
Ti
titanium
(solid) | 23
V
vanadium
(solid) | 24
Cr
chromium
(solid) | 25
Mn
manganese
(solid) | 26
Fe
iron
(solid) | 27
Co
cobalt
(solid) | 28
Ni
nickel
(solid) | 29
Cu
copper
(solid) | 30
Zn
zinc
(solid) | 31
Ga
gallium
(solid) | 32
Ge
germanium
(solid) | 33
As
arsenic
(solid) | 34
Se
selenium
(solid) | 35
Br
bromine
(liquid) | 36
Kr
krypton
(gas) | |
| 37
Rb
rubidium
(solid) | 38
Sr
strontium
(solid) | 39
Y
yttrium
(solid) | 40
Zr
zirconium
(solid) | 41
Nb
niobium
(solid) | 42
Mo
molybdenum
(solid) | 43
Tc
technetium
(solid) | 44
Ru
ruthenium
(solid) | 45
Rh
rhodium
(solid) | 46
Pd
palladium
(solid) | 47
Ag
silver
(solid) | 48
Cd
cadmium
(solid) | 49
In
indium
(solid) | 50
Sn
tin
(solid) | 51
Sb
antimony
(solid) | 52
Te
tellurium
(solid) | 53
I
iodine
(solid) | 54
Xe
xenon
(gas) | |
| 55
Cs
cesium
(solid) | 56
Ba
barium
(solid) | 57-71
lanthanoids | 72
Hf
hafnium
(solid) | 73
Ta
tantalum
(solid) | 74
W
tungsten
(solid) | 75
Re
rhenium
(solid) | 76
Os
osmium
(solid) | 77
Ir
iridium
(solid) | 78
Pt
platinum
(solid) | 79
Au
gold
(solid) | 80
Hg
mercury
(liquid) | 81
Tl
thallium
(solid) | 82
Pb
lead
(solid) | 83
Bi
bismuth
(solid) | 84
Po
polonium
(solid) | 85
At
astatine
(solid) | 86
Rn
radon
(gas) | |
| 87
Fr
francium
(solid) | 88
Ra
radium
(solid) | 89-103
actinoids | 104
Rf
rutherfordium
(solid) | 105
Db
dubnium
(solid) | 106
Sg
seaborgium
(solid) | 107
Bh
bohrium
(solid) | 108
Hs
hassium
(solid) | 109
Mt
meitnerium
(solid) | 110
Ds
darmstadtium
(solid) | 111
Rg
roentgenium
(solid) | 112
Cn
copernicium
(solid) | 113
Nh
nihonium
(solid) | 114
Fl
flerovium
(solid) | 115
Mc
moscovium
(solid) | 116
Lv
livermorium
(solid) | 117
Ts
tennessine
(solid) | 118
Og
oganeson
(solid) | |

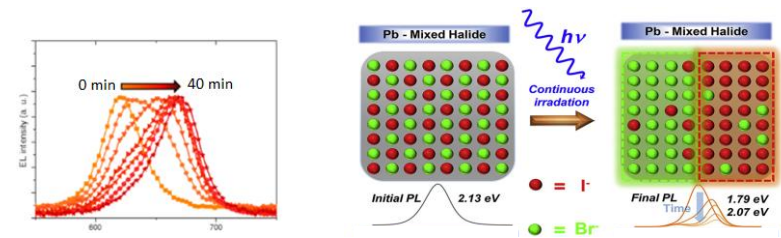
35

Halide ion migration

• While perovskite has a fixed structure, its ions are partially quite mobile...

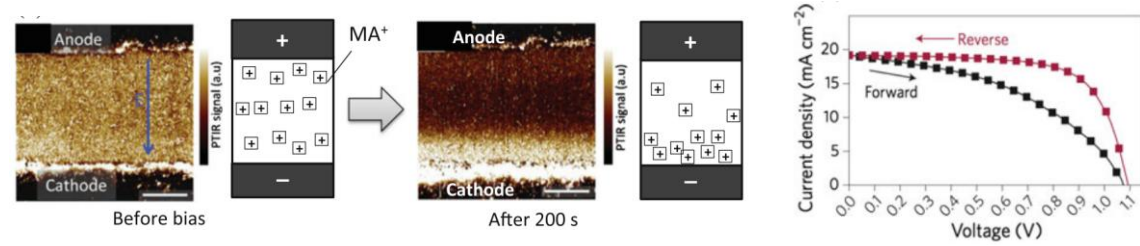


• In mixed halide (e.g. Br/I) LED, the color changes with time...

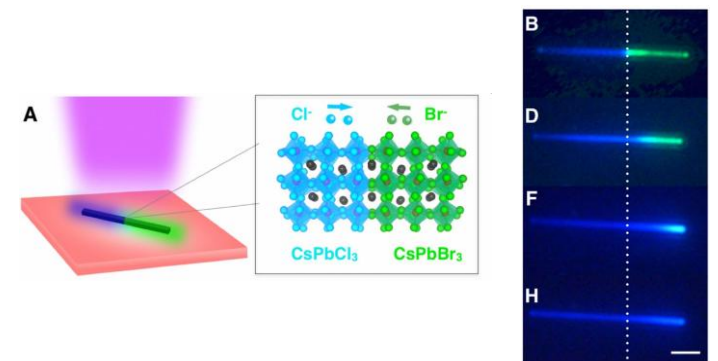
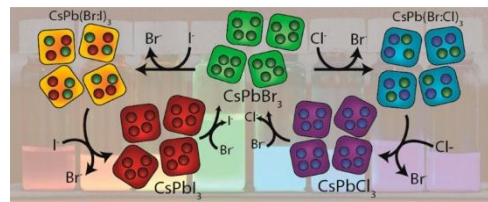
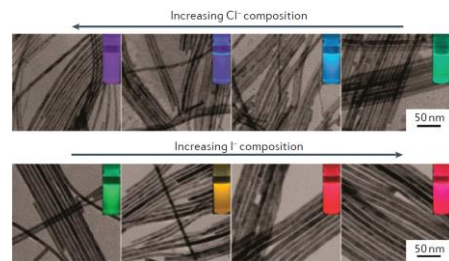


• The halide ions are moving and segregating into separate phases

• Also problematic in solar cells...
(**hysteresis**)



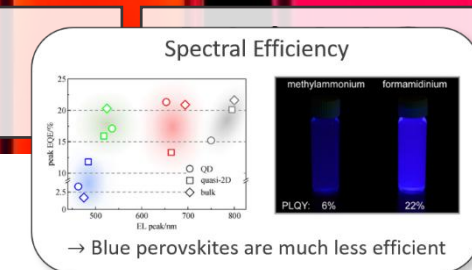
• However, can also be useful...
(**anion exchange**)



D. Hong et al. *iScience* 8, 101415 (2020)
M. Lai et al. *PNAS* 115, 11929-11934 (2018)
D. Zhang et al. *JACS* 238, 7236-7239 (2016)
Q.A. Akkermann et al. *JACS* 137, 10276-10281 (2015)

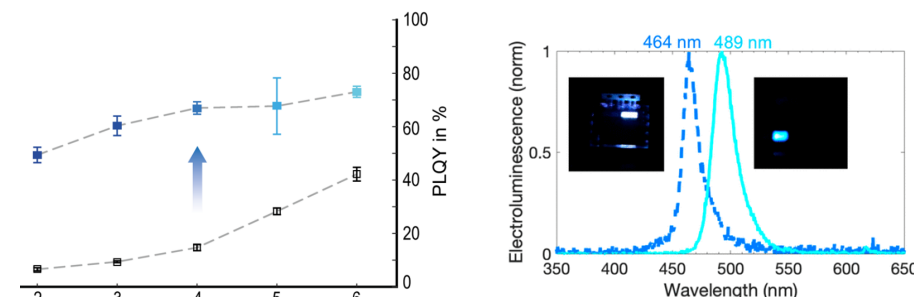
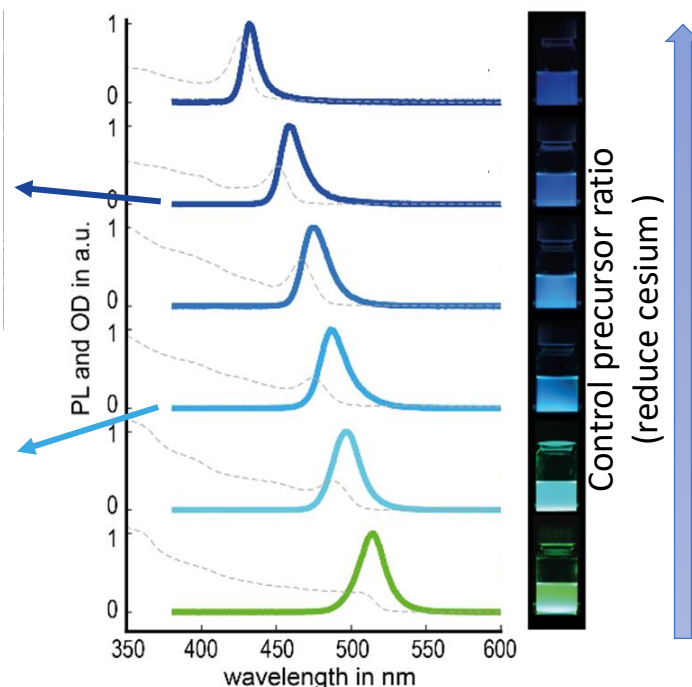
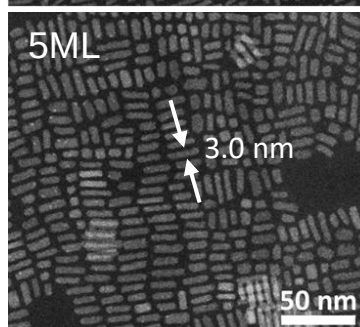
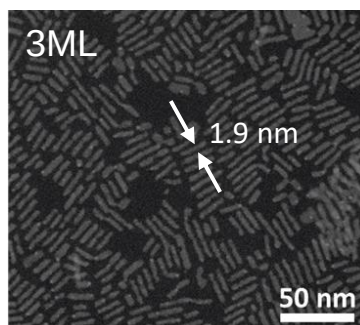
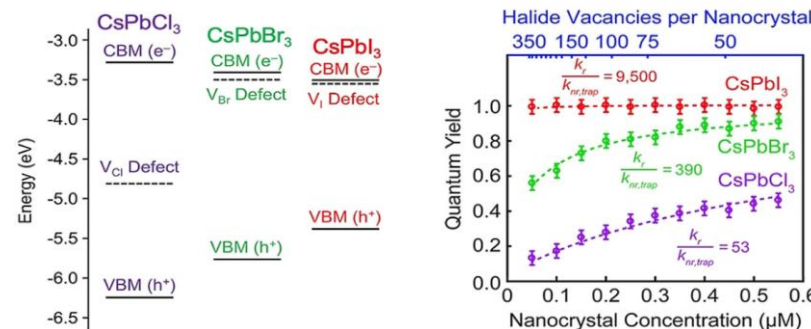
Improving perovskites in the blue spectral range

Blue emitting perovskites perform far worse than all other colors



While iodide- and bromide perovskites are defect tolerant chloride perovskites are not (chloride vacancies)

Alternative: Utilize quantum confinement



- Quantum yields up to 80% (new data)
- Blue LEDs realized
- Quantum confinement: exciton orientation
→ (outcoupling enhanced)